

# Tema 3



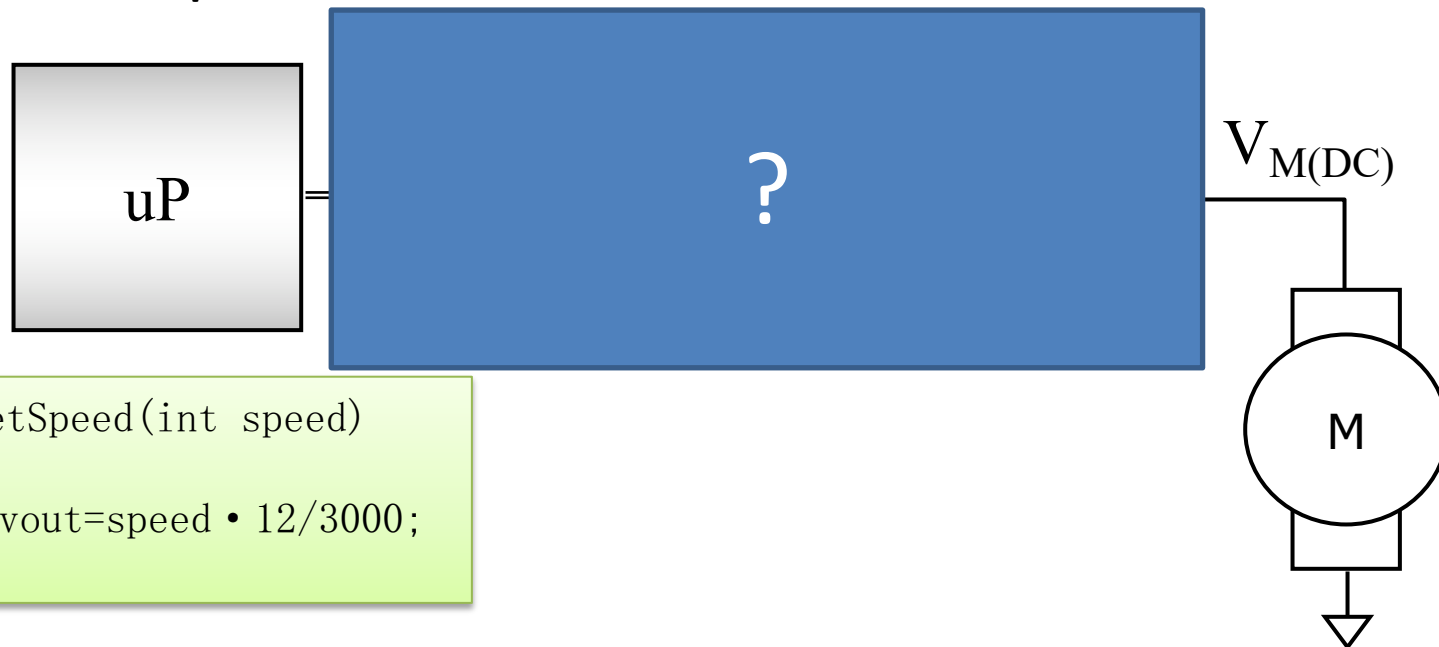
Convertidores Digitales Analógicos  
(DAC)

# Índice

1. Introducción
2. Especificaciones estáticas
3. Especificaciones dinámicas
4. Técnicas de conversión
5. Ejemplos: DAC08 y TLC5620C

# Aplicación

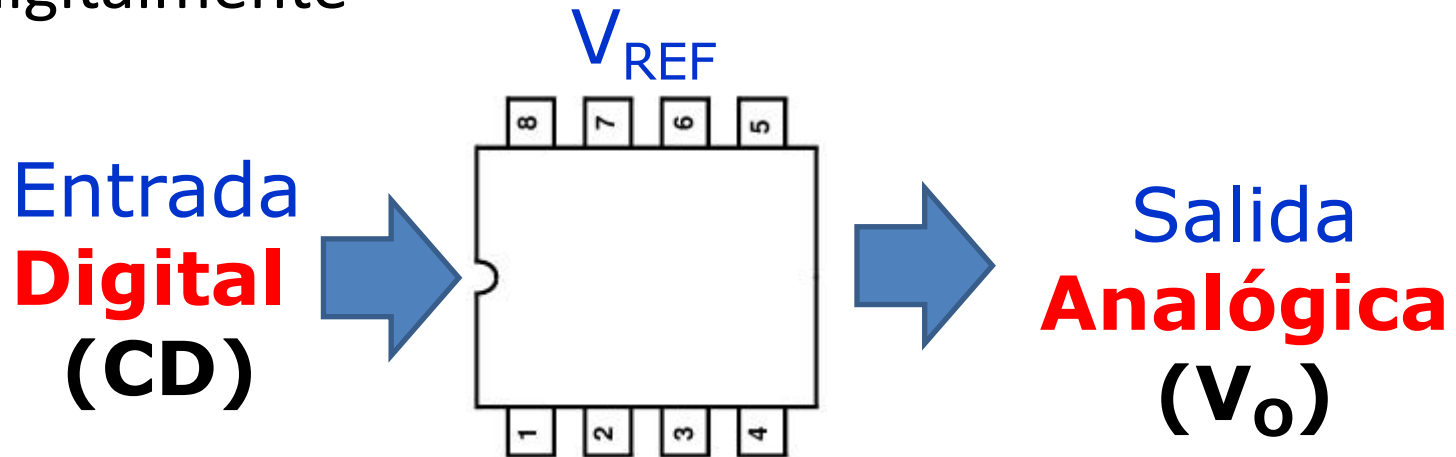
- ❑ Un motor de continua gira a una velocidad proporcional a la tensión aplicada en sus extremos
  - Por ejemplo: el motor utilizado en Sistemas de Control 0V está parado, y con 12V gira a 3000rpm
- ❑ ¿Cómo conseguimos fijar su velocidad desde un microprocesador?



```
void setSpeed(int speed)
{
    set vout=speed • 12/3000;
}
```

# Funcionamiento

- ❑ Un DAC es un dispositivo que permite generar una tensión analógica, cuyo valor se puede controlar digitalmente



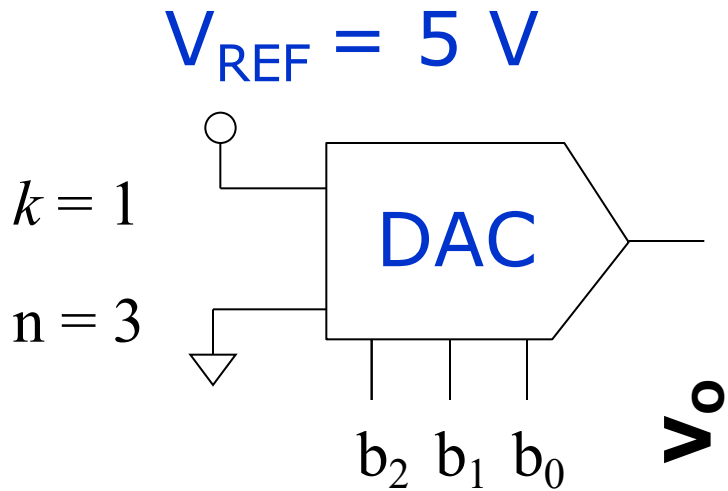
$$V_0 = k \cdot V_{REF} \frac{CD}{2^n}$$

$V_{FSR}$ : Rango de Fondo de Escala<sup>4</sup>



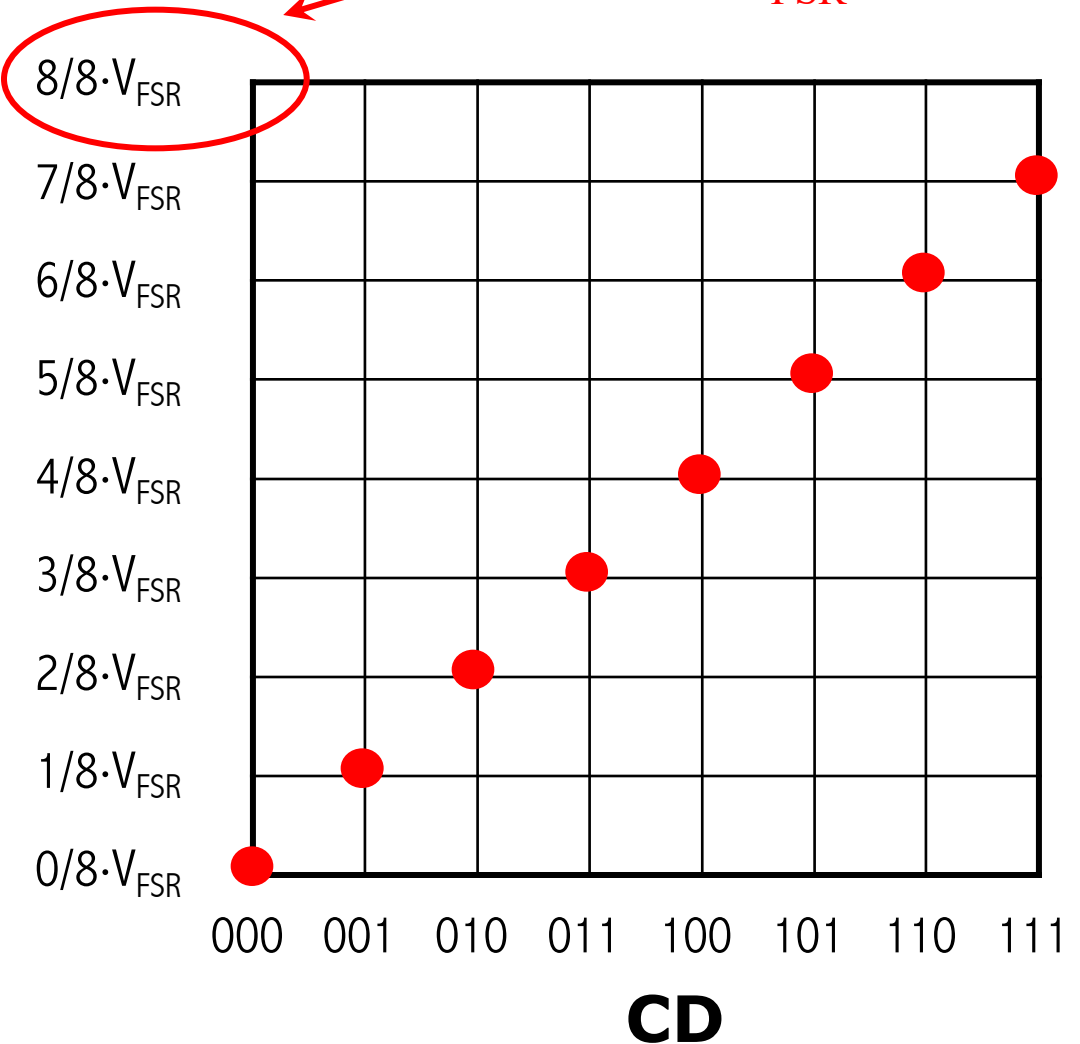
# Ejemplo

Nunca se alcanza el  $V_{FSR}$  !



$$V_o = k \cdot V_{REF} \frac{CD}{2^n}$$

$$V_{FSR} = k \cdot V_{REF}$$

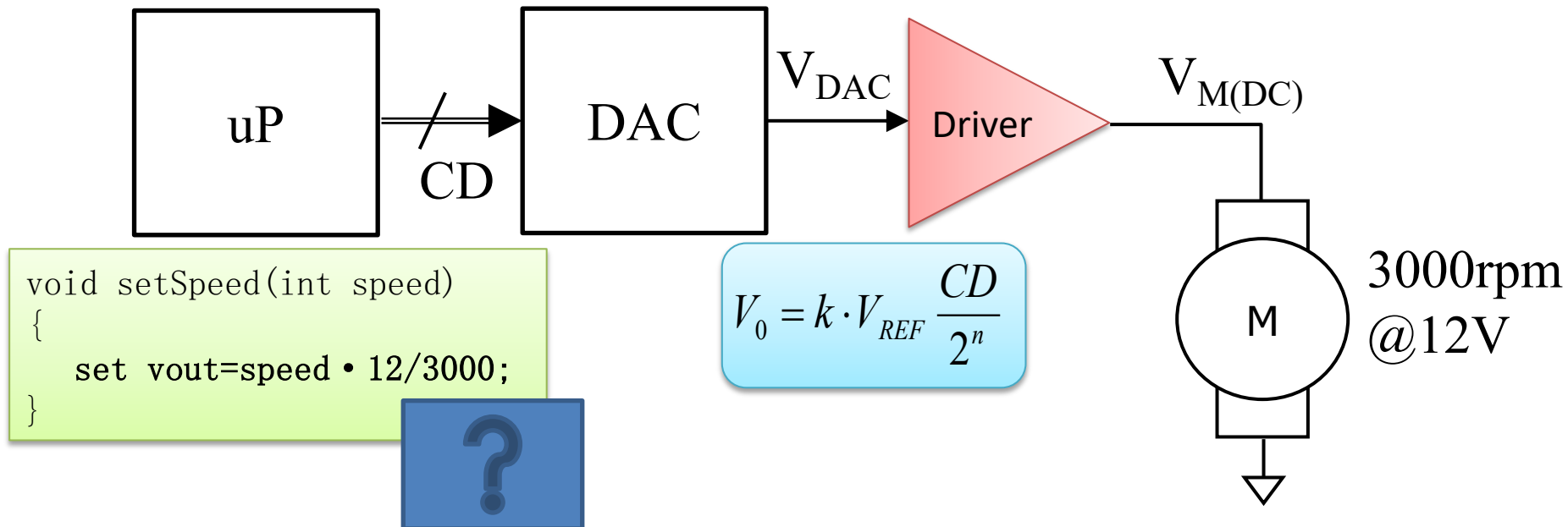


# ¿Porqué hace falta el “driver”?

## ❑ Dos funciones:

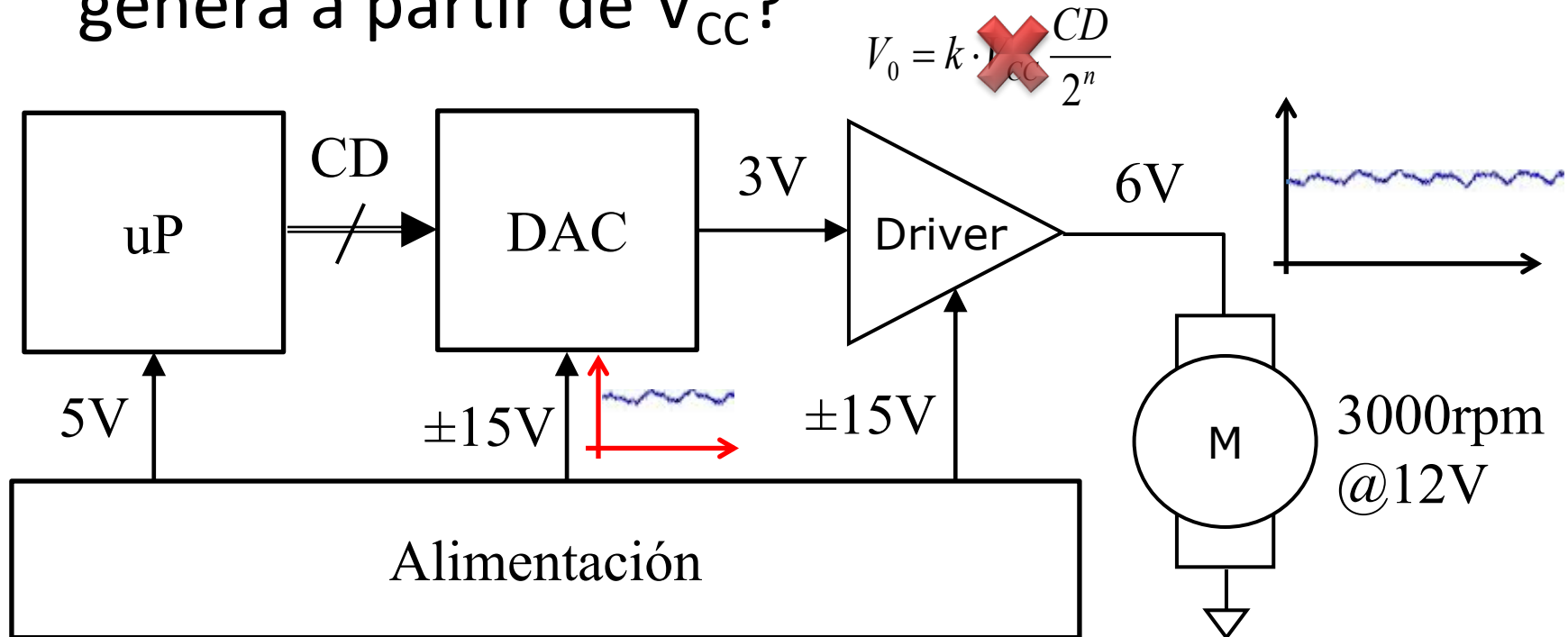
- Adaptar los niveles de tensión de salida del DAC
- Proporcionar corriente suficiente al motor

## ❑ Ojo, tiene implicaciones!



# ¿Por qué $V_{REF}$ ?

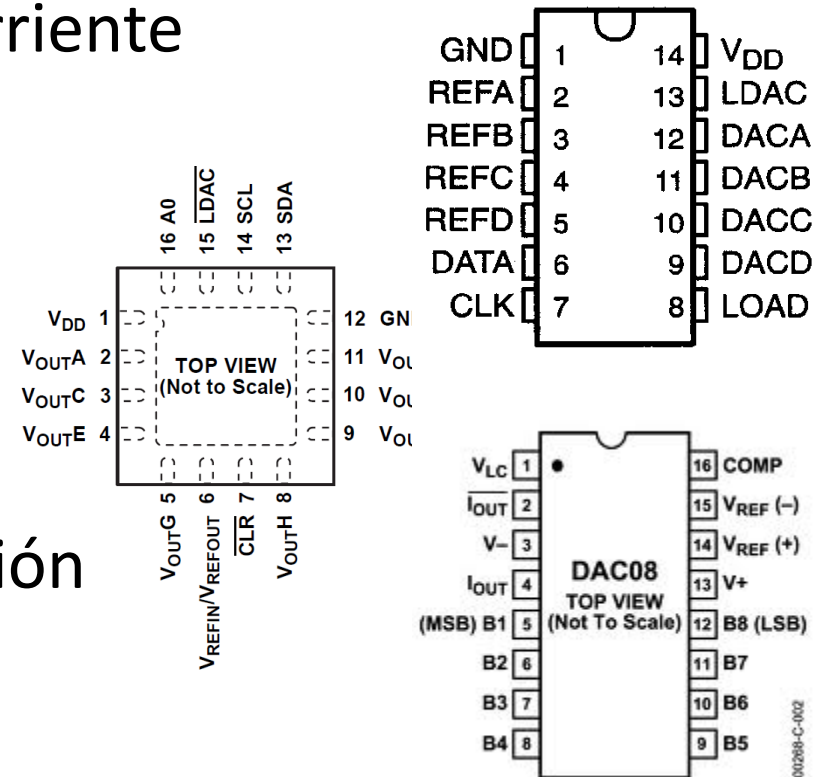
- ❑ Si fijo un CD para que el motor gire a 1500rpm, la velocidad del motor no debería cambiar
- ❑ ¿Puedo garantizar esto si la  $V_O$  del DAC se genera a partir de  $V_{CC}$ ?



# Consideraciones

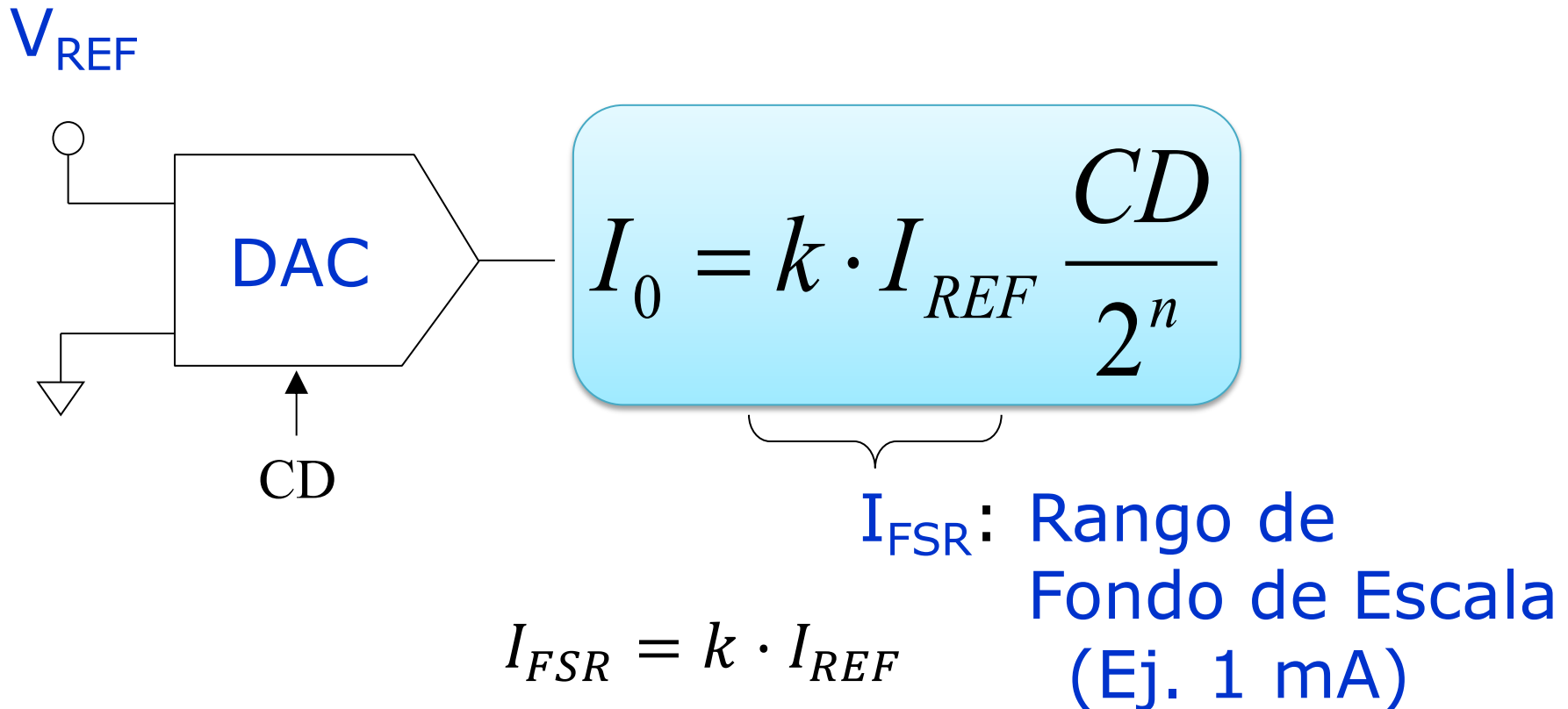
❑ Cada modelo tiene sus peculiaridades:

- Interfaz digital: serie (I2C, SPI...), paralelo
- Tipo salida: tensión, corriente
- Número de salidas
- Tensión de referencia
- Polaridad
- Rango
- Tensiones de alimentación
- Terminales de ajuste
- Encapsulado

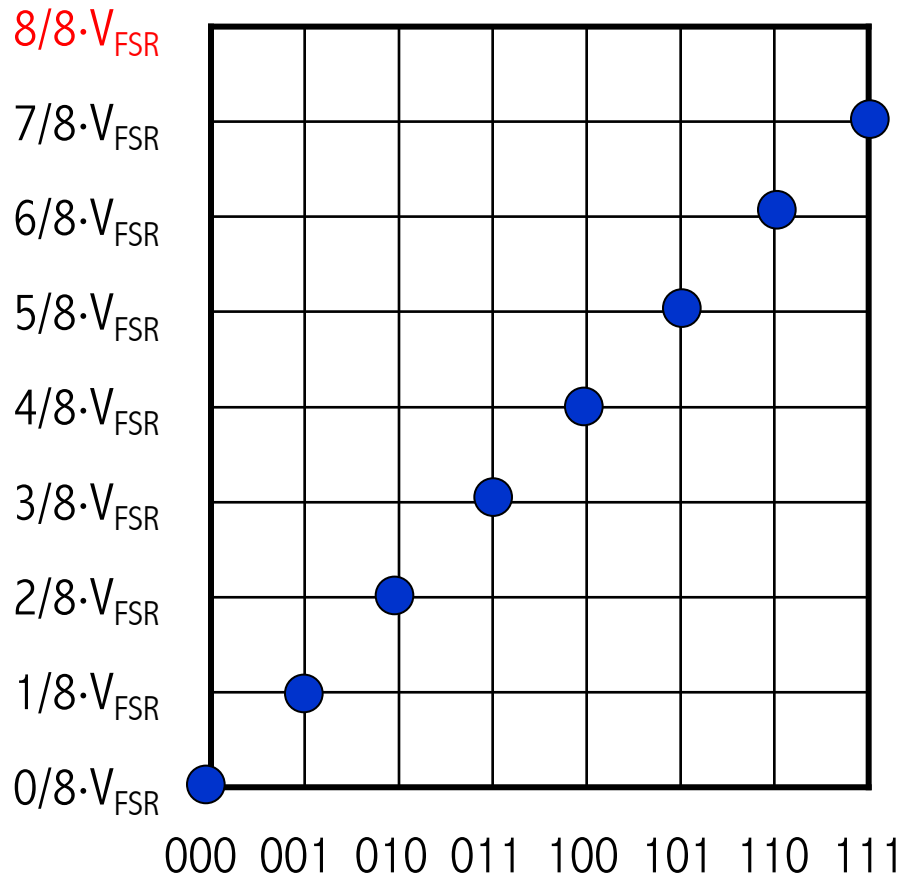


# Salida en corriente

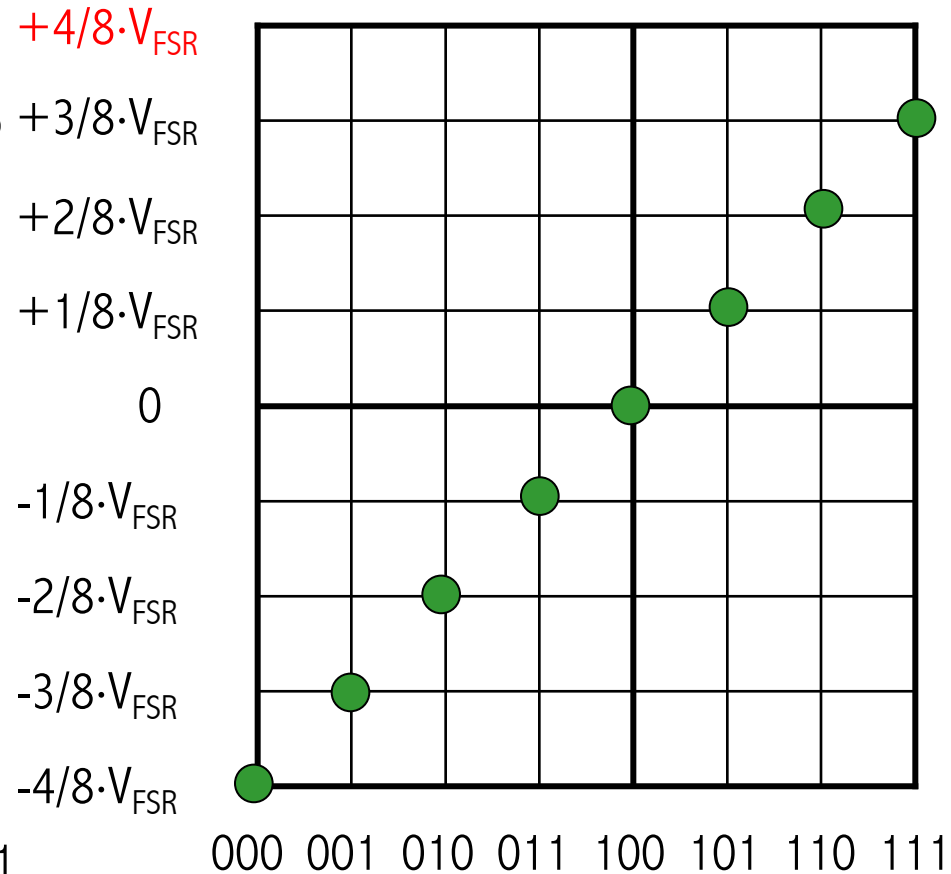
- Normalmente la  $V_{REF}$  se convierte internamente en una  $I_{REF}$  a partir de la cual se genera  $I_O$



# Característica de transferencia



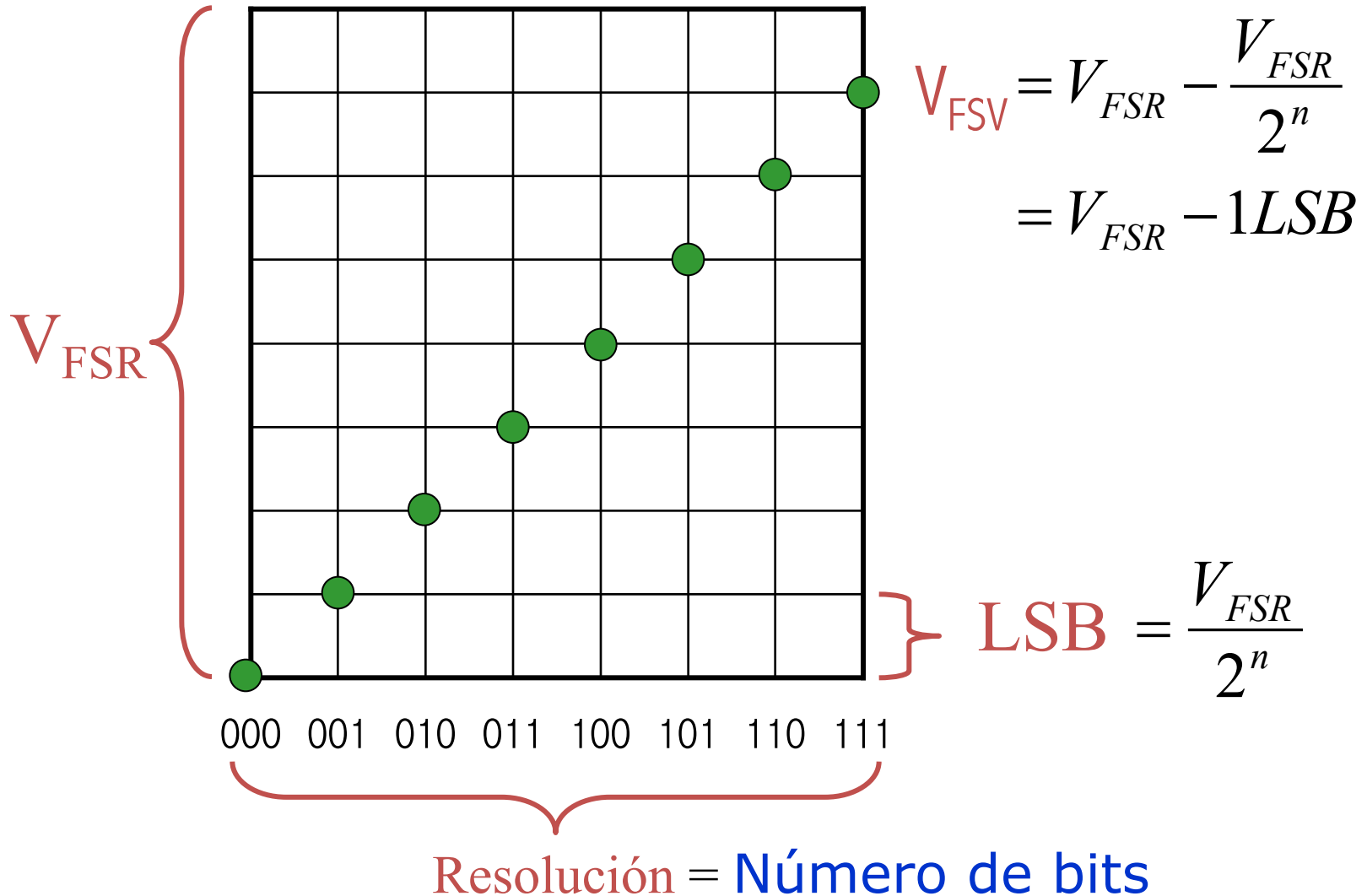
UNIPOLAR



BIPOLAR

$$V_{0(Bip)} = V_{0(Uni)} - \frac{k \cdot V_{REF}}{2}$$

# Definiciones



# LSB (FSR=10V)

| RESOLUTION<br>N | $2^N$ | VOLTAGE | ppm FS | % FS | dB FS |
|-----------------|-------|---------|--------|------|-------|
| 8-bit           | 256   | 39.1 mV | 3,906  | 0.39 | -48   |



# LSB (FSR=10V)

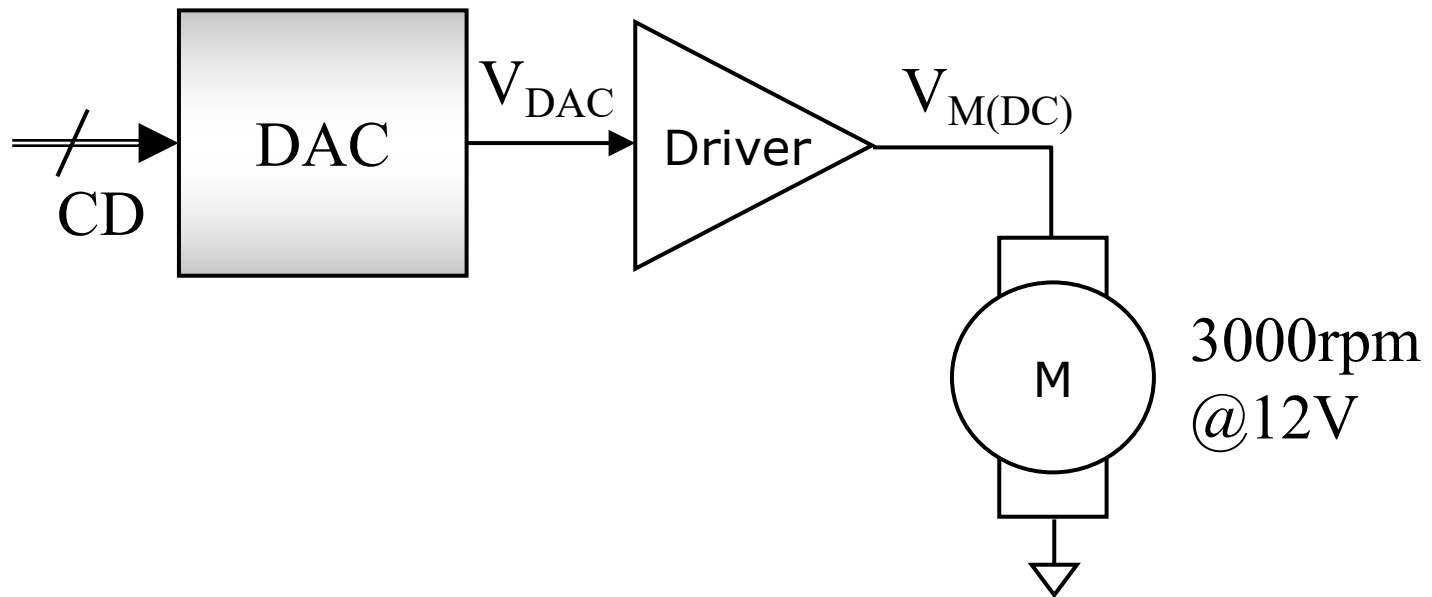
| RESOLUTION<br>N | $2^N$      | VOLTAGE                   | ppm FS  | % FS     | dB FS |
|-----------------|------------|---------------------------|---------|----------|-------|
| 2-bit           | 4          | 2.5 V                     | 250,000 | 25       | -12   |
| 4-bit           | 16         | 625 mV                    | 62,500  | 6.25     | -24   |
| 6-bit           | 64         | 156 mV                    | 15,625  | 1.56     | -36   |
| 8-bit           | 256        | 39.1 mV                   | 3,906   | 0.39     | -48   |
| 10-bit          | 1,024      | 9.77 mV (10 mV)           | 977     | 0.098    | -60   |
| 12-bit          | 4,096      | 2.44 mV                   | 244     | 0.024    | -72   |
| 14-bit          | 16,384     | 610 $\mu$ V               | 61      | 0.0061   | -84   |
| 16-bit          | 65,536     | 153 $\mu$ V               | 15      | 0.0015   | -96   |
| 18-bit          | 262,144    | 38 $\mu$ V                | 4       | 0.0004   | -108  |
| 20-bit          | 1,048,576  | 9.54 $\mu$ V (10 $\mu$ V) | 1       | 0.0001   | -120  |
| 22-bit          | 4,194,304  | 2.38 $\mu$ V              | 0.24    | 0.000024 | -132  |
| 24-bit          | 16,777,216 | 596 nV*                   | 0.06    | 0.000006 | -144  |

# Índice

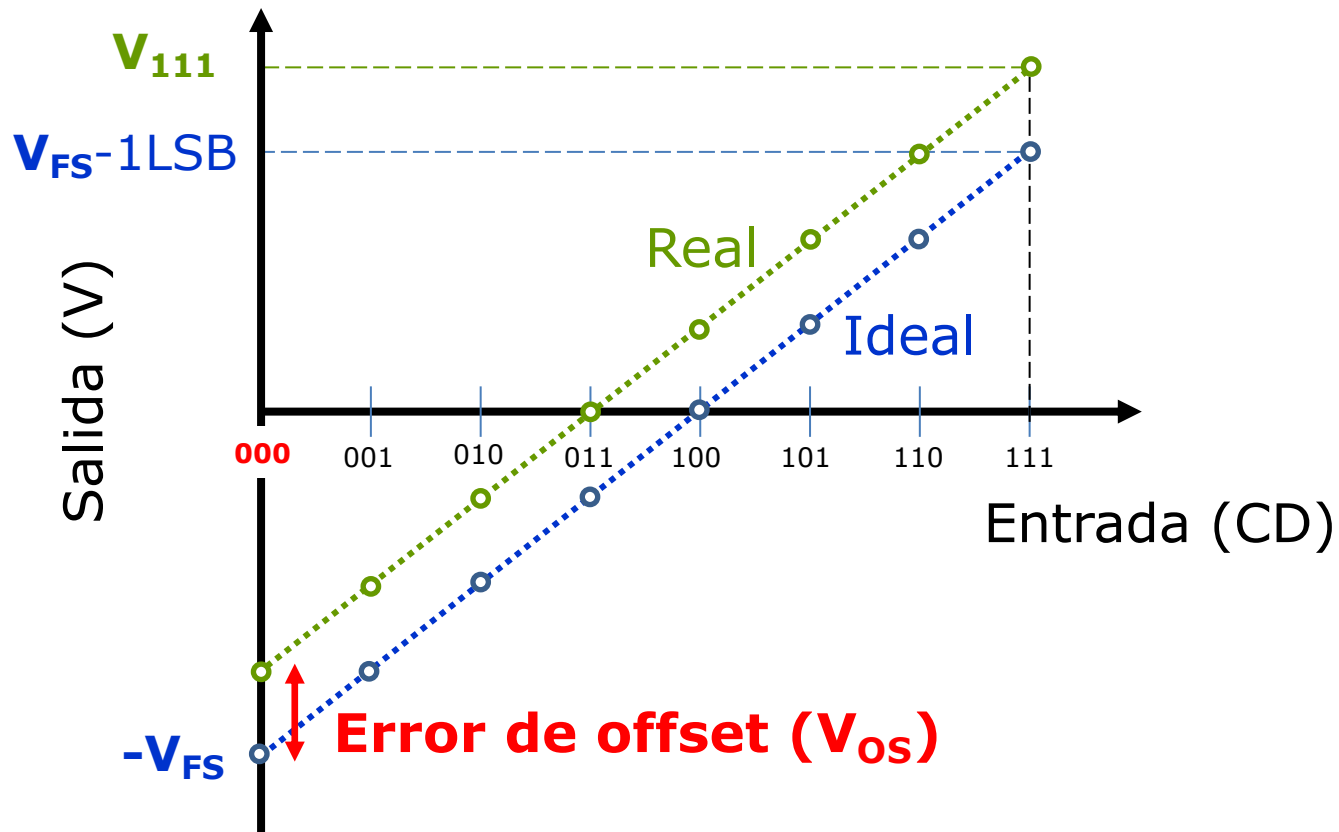
1. Introducción
- 2. Especificaciones estáticas**
3. Especificaciones dinámicas
4. Técnicas de conversión
5. Ejemplos: DAC08 y TLC5620C

# Funcionamiento real en estático

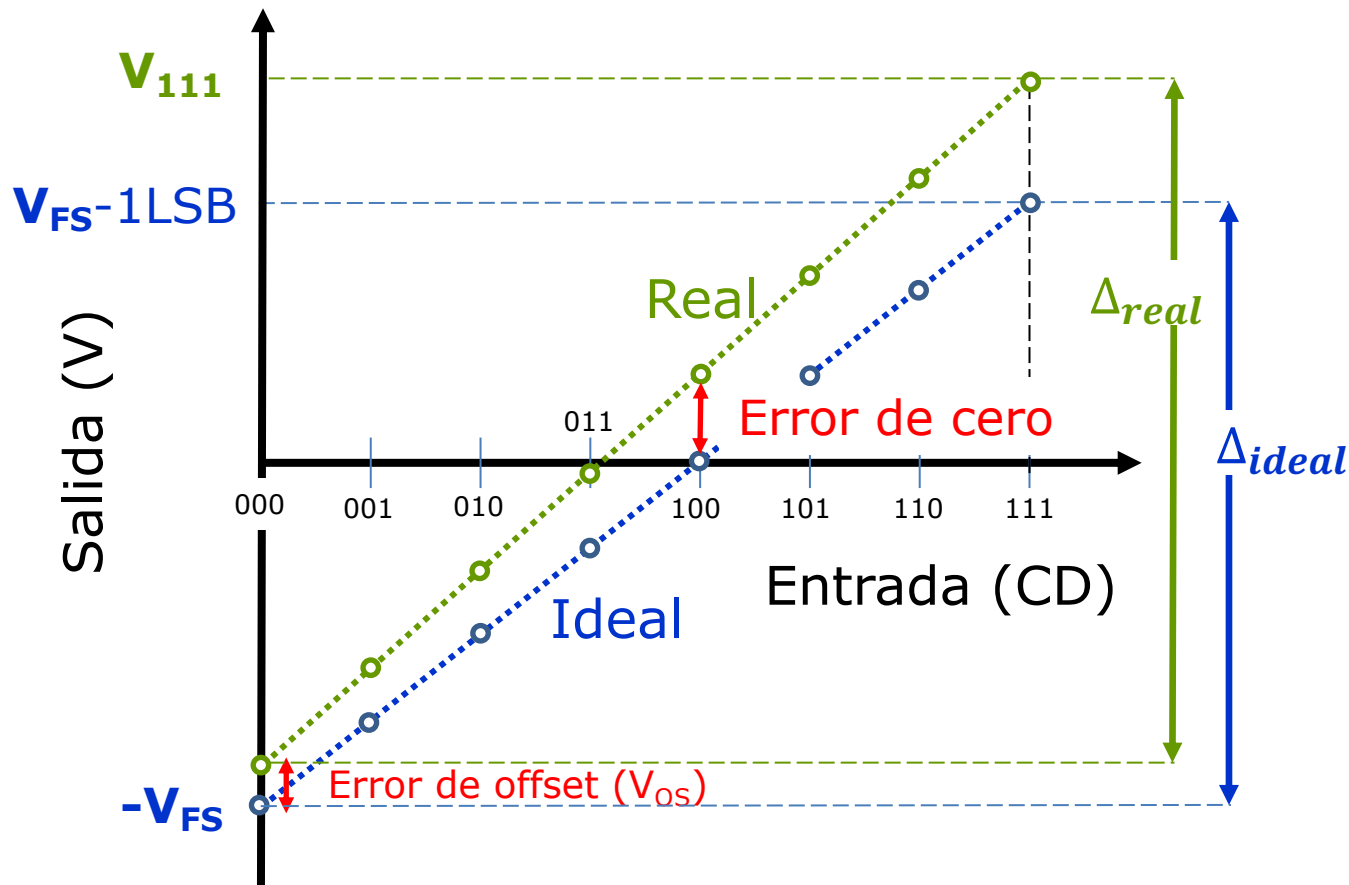
- ❑ Ejemplo: con DAC  $V_{FSR}=5V$  y  $n=8$  bits controlar un motor de DC 3000rpm@12V
  - ¿A qué velocidad girará el motor con  $CD=0x00$ ??
  - ¿Y con el  $0x1F$ ?
  - ¿Y con el  $0xFF$ ?



# Error de offset

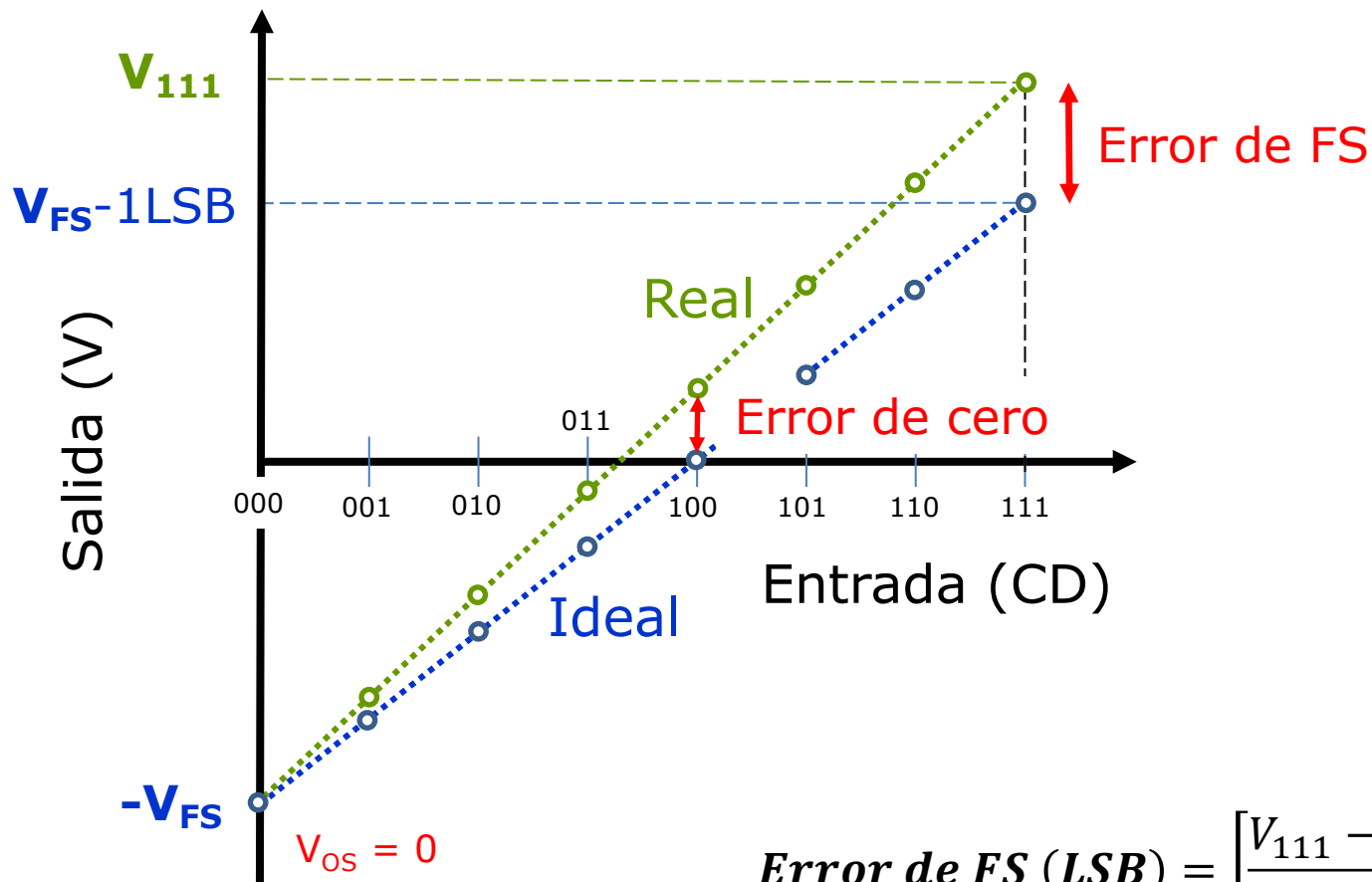


# Error de Ganancia



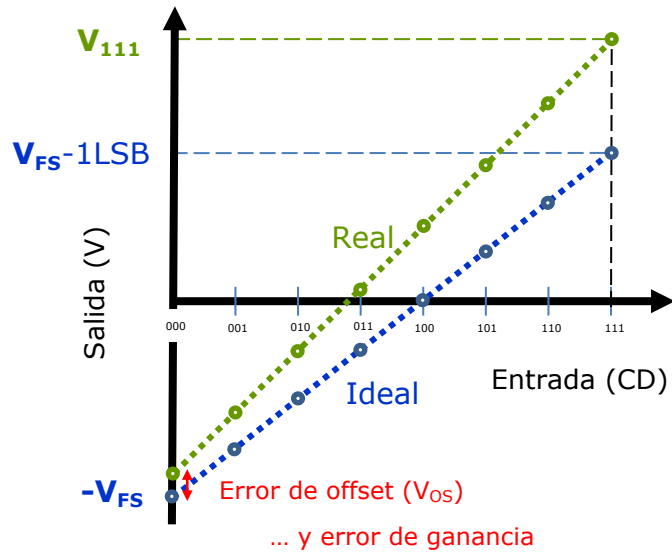
$$\text{Error de ganancia (\%)} = \left[ \frac{\Delta_{real} - \Delta_{ideal}}{\Delta_{ideal}} \right] \cdot 100 = \left[ \frac{V_{111} + V_{FS} - V_{OS}}{2V_{FS} - 1 \text{ LSB}} - 1 \right] \cdot 100$$

# Error de Ganancia (Error de Fondo de Escala)

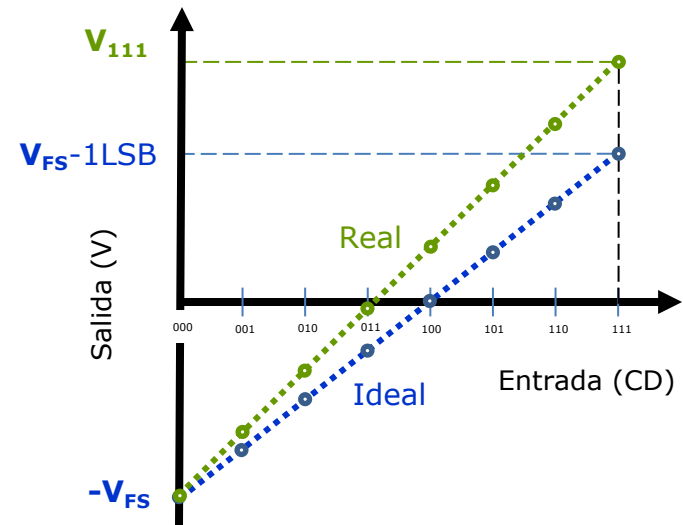
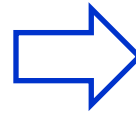


$$Error\ de\ FS\ (LSB) = \left[ \frac{V_{111} - (V_{FS} - 1LSB)}{1\ LSB} \right]$$

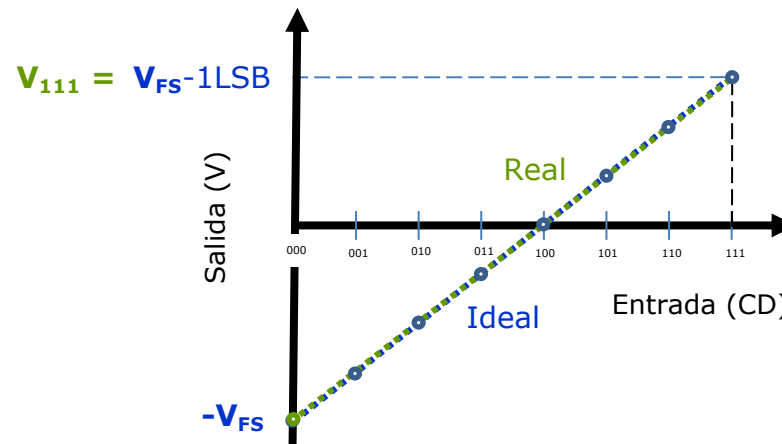
# Ajuste de los errores de offset y ganancia



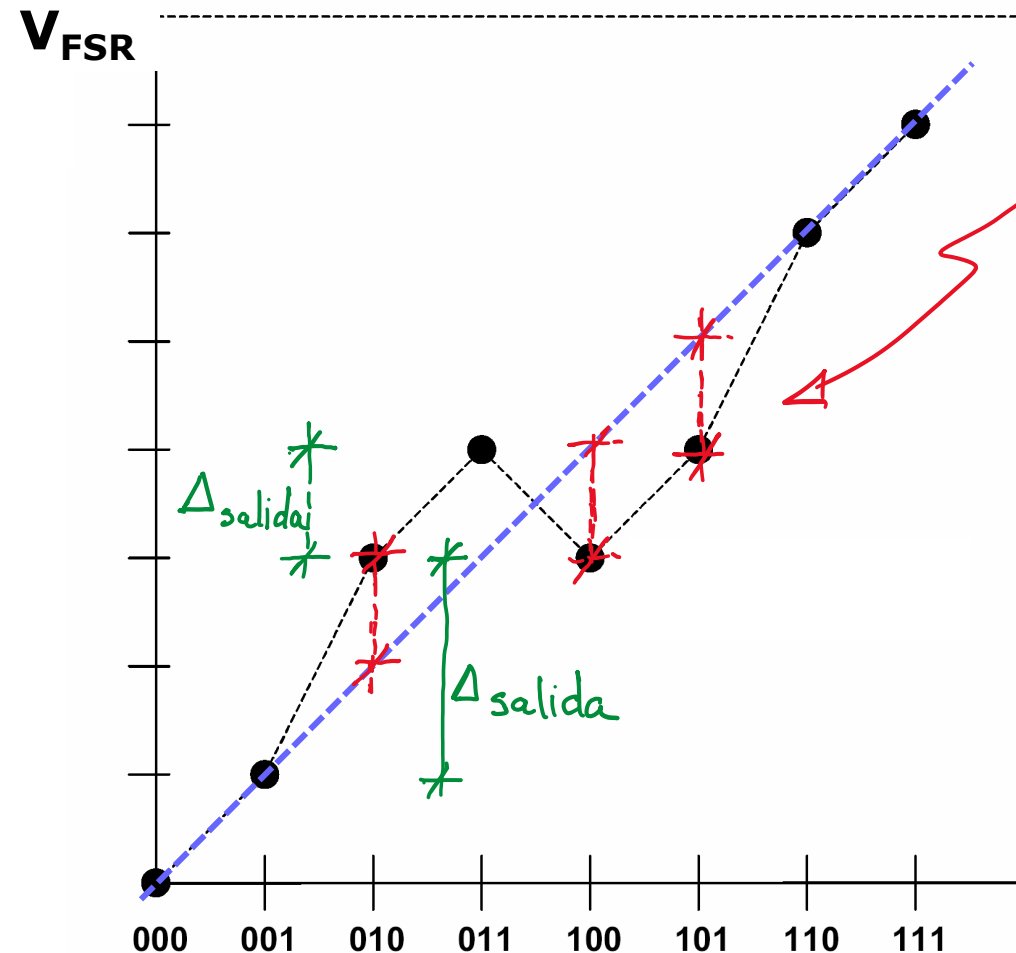
1º Corrección  
de offset



2º Corrección  
de ganancia



# No linealidades de la característica de transferencia real



**INL:** desviación máxima del valor ideal (una vez ajustados offset y ganancia)

**DNL:** desviación máxima del ancho del escalón respecto su valor ideal (1 LSB)

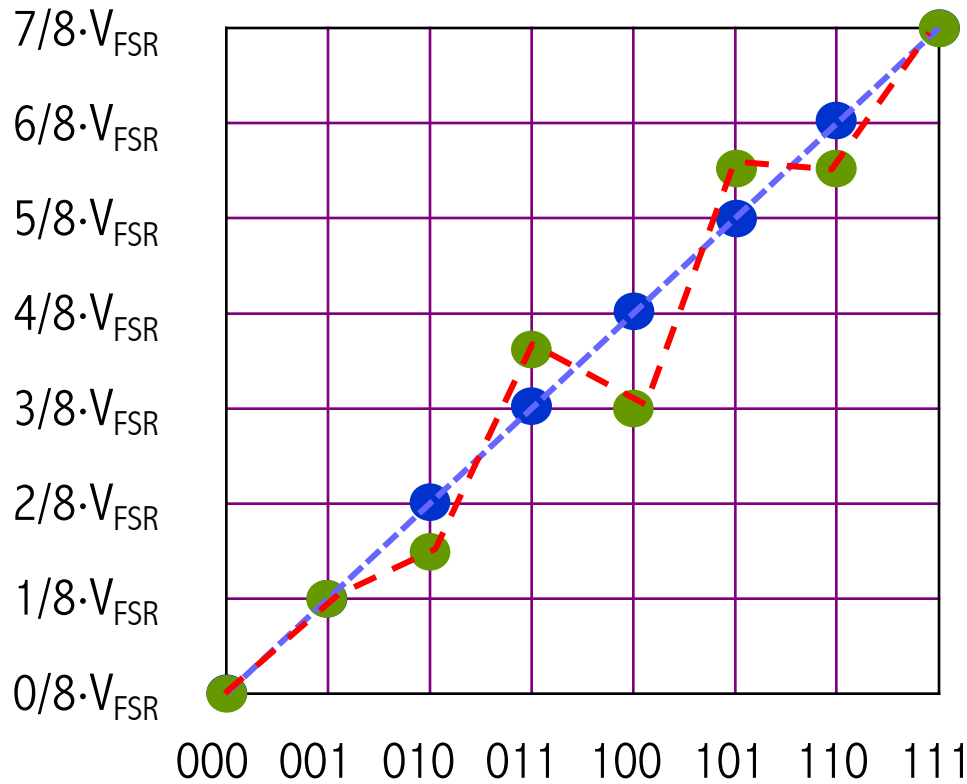
$$DNL = [\Delta_{salida}(LSB) - 1LSB]_{\max}$$



# Ejemplo

**INL:** desviación máxima del valor ideal (una vez ajustados offset y ganancia)

**DNL:** desviación máxima del ancho del escalón respecto su valor ideal



| Código | INL | DNL |
|--------|-----|-----|
| 000    |     |     |
| 001    |     |     |
| 010    |     |     |
| 011    |     |     |
| 100    |     |     |
| 101    |     |     |
| 110    |     |     |
| 111    |     |     |

**INL =**

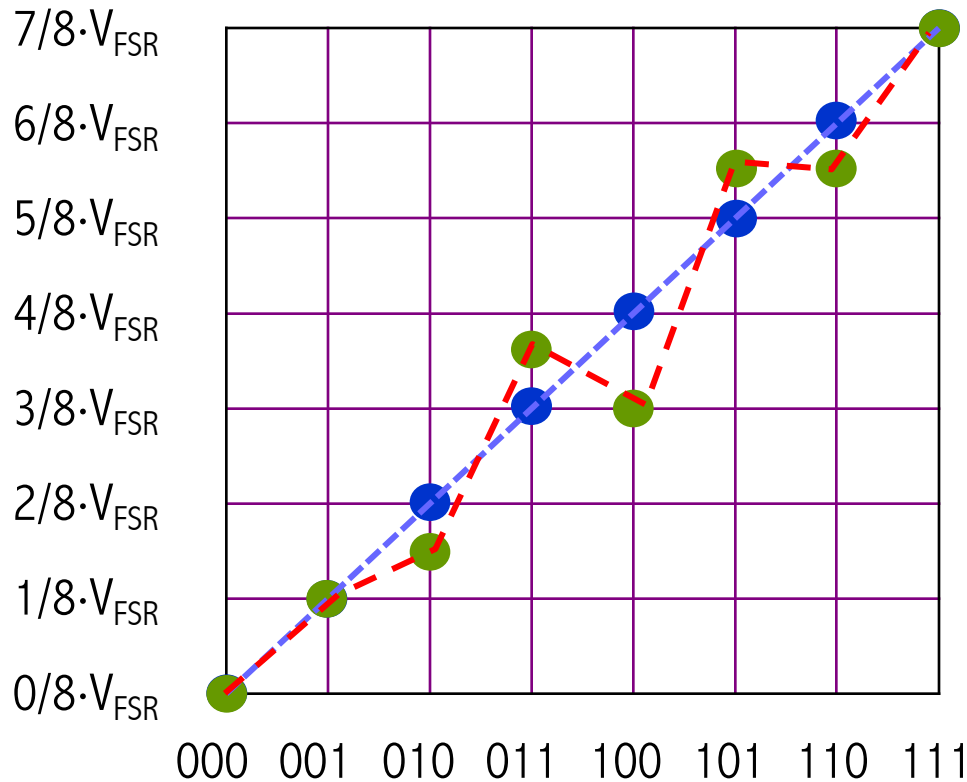
$$DNL = [\Delta_{salida}(LSB) - 1LSB]_{max}$$

**DNL =**

# Ejemplo

**INL:** desviación máxima del valor ideal (una vez ajustados offset y ganancia)

**DNL:** desviación máxima del ancho del escalón respecto su valor ideal



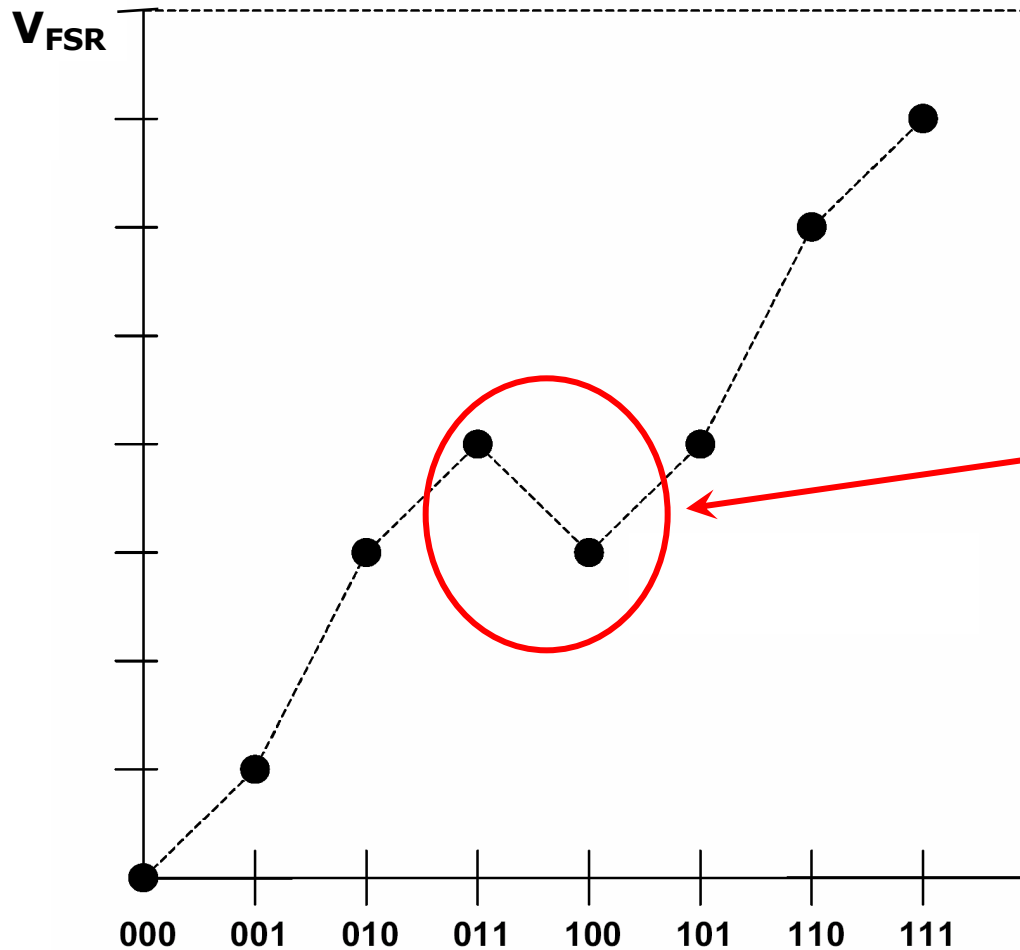
| Código | INL  | DNL  |
|--------|------|------|
| 000    | 0    | 0    |
| 001    | 0    | 0    |
| 010    | -0.5 | -0.5 |
| 011    | +0.5 | 1    |
| 100    | -1   | -1.5 |
| 101    | +0.5 | +1.5 |
| 110    | -0.5 | -1   |
| 111    | 0    | 0.5  |

$$\text{INL} = 1 \text{ LSB}$$

$$\text{DNL} = [\Delta_{\text{salida}}(\text{LSB}) - 1\text{LSB}]_{\text{max}}$$

$$\longrightarrow \text{DNL} = 1.5 \text{ LSB}$$

# Monotonicidad



Un DAC es MONÓTONO si la pendiente NO cambia de signo

**DNL < -1.0 LSB**

La tensión de salida **disminuye** cuando **aumenta** el código aplicado en la entrada. La pendiente cambia, es **NO MONÓTONO**

# Ejemplo errores DAC

| N      | 3                  | VFSR                | 5                                | LSB                                            | 0,625  |          |        |          |
|--------|--------------------|---------------------|----------------------------------|------------------------------------------------|--------|----------|--------|----------|
| Código | $V_{DAC}$<br>Ideal | $V_{DAC}$<br>Medida | $V_{DAC}$<br>Offset<br>corregido | $V_{DAC}$<br>Offset y<br>Ganancia<br>corregida | INL(V) | INL(LSB) | DNL(V) | DNL(LSB) |
| 0      | -2,500             | -2,400              |                                  |                                                |        |          |        |          |
| 1      | -1,875             | -1,700              |                                  |                                                |        |          |        |          |
| 2      | -1,250             | -1,000              |                                  |                                                |        |          |        |          |
| 3      | -0,625             | 0,200               |                                  |                                                |        |          |        |          |
| 4      | 0,000              | -0,120              |                                  |                                                |        |          |        |          |
| 5      | 0,625              | 1,000               |                                  |                                                |        |          |        |          |
| 6      | 1,250              | 0,955               |                                  |                                                |        |          |        |          |
| 7      | 1,875              | 1,523               |                                  |                                                |        |          |        |          |

|                 | V | LSB | V/código |
|-----------------|---|-----|----------|
| Error de offset |   |     |          |
| Error de cero   |   |     |          |
| Error Ganancia  |   |     |          |
| INL             |   |     |          |
| DNL             |   |     |          |

# Ejemplo errores DAC: corrección de offset

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido |
|--------|---------------------------|----------------------------|-----------------------------------------|
| 0      | -2,500                    | -2,400                     |                                         |
| 1      | -1,875                    | -1,700                     |                                         |
| 2      | -1,250                    | -1,000                     |                                         |
| 3      | -0,625                    | 0,200                      |                                         |
| 4      | 0,000                     | -0,120                     |                                         |
| 5      | 0,625                     | 1,000                      |                                         |
| 6      | 1,250                     | 0,955                      |                                         |
| 7      | 1,875                     | 1,523                      |                                         |

Error de offset:

$$V_{OS} = V_{real(000)} - V_{ideal(000)}$$

$$V_{OS} =$$

Error de offset corregido:

$$V_{offset\_corregido} = V_{real} - V_{OS}$$

|                 | V | LSB |  |
|-----------------|---|-----|--|
| Error de offset |   |     |  |
| Error de cero   |   |     |  |

# Ejemplo errores DAC: corrección de offset

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido |
|--------|---------------------------|----------------------------|-----------------------------------------|
| 0      | -2,500                    | -2,400                     | -2,500                                  |
| 1      | -1,875                    | -1,700                     | -1,800                                  |
| 2      | -1,250                    | -1,000                     | -1,100                                  |
| 3      | -0,625                    | 0,200                      | 0,100                                   |
| 4      | 0,000                     | -0,120                     | -0,220                                  |
| 5      | 0,625                     | 1,000                      | 0,900                                   |
| 6      | 1,250                     | 0,955                      | 0,855                                   |
| 7      | 1,875                     | 1,523                      | 1,423                                   |

Error de offset:

$$V_{OS} = V_{real(000)} - V_{ideal(000)}$$

$$V_{OS} = 0,1$$

Error de offset corregido:

$$V_{offset\_corregido} = V_{real} - V_{OS}$$

|                 | V      | LSB    |  |
|-----------------|--------|--------|--|
| Error de offset | 0,100  | 0,160  |  |
| Error de cero   | -0,120 | -0,192 |  |

# Ejemplo errores DAC: corrección de ganancia

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|
| 0      | -2,500                    | -2,400                     | -2,500                                  |                                                       |
| 1      | -1,875                    | -1,700                     | -1,800                                  |                                                       |
| 2      | -1,250                    | -1,000                     | -1,100                                  |                                                       |
| 3      | -0,625                    | 0,200                      | 0,100                                   |                                                       |
| 4      | 0,000                     | -0,120                     | -0,220                                  |                                                       |
| 5      | 0,625                     | 1,000                      | 0,900                                   |                                                       |
| 6      | 1,250                     | 0,955                      | 0,855                                   |                                                       |
| 7      | 1,875                     | 1,523                      | 1,423                                   |                                                       |

Error de ganancia:

$$EG = V_{real\ (off-corr)_{(111)}} - V_{ideal_{(111)}}$$

$$EG =$$

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  |        |        |          |

Error de offset y ganancia corregidos:

$$V_{off\_corr, gain\_corr} = V_{o\_c} - \left( EG \cdot \frac{VDC}{2^n - 1} \right)$$

$$(V/código) = \frac{EG}{2^n - 1}$$

# Ejemplo errores DAC: corrección de ganancia

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|
| 0      | -2,500                    | -2,400                     | -2,500                                  | -2,500                                                |
| 1      | -1,875                    | -1,700                     | -1,800                                  | -1,735                                                |
| 2      | -1,250                    | -1,000                     | -1,100                                  | -0,971                                                |
| 3      | -0,625                    | 0,200                      | 0,100                                   | 0,294                                                 |
| 4      | 0,000                     | -0,120                     | -0,220                                  | 0,038                                                 |
| 5      | 0,625                     | 1,000                      | 0,900                                   | 1,223                                                 |
| 6      | 1,250                     | 0,955                      | 0,855                                   | 1,242                                                 |
| 7      | 1,875                     | 1,523                      | 1,423                                   | 1,875                                                 |

Error de ganancia:

$$EG = V_{real\ (off-corr)_{(111)}} - V_{ideal_{(111)}}$$

$$EG = -0,452$$

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  | -0,452 | -0,723 | -0,06457 |

Error de offset y ganancia corregidos:

$$V_{off\_corr, gain\_corr} = V_{o\_c} - \left( EG \cdot \frac{VDC}{2^n - 1} \right)$$

$$(V/código) = \frac{EG}{2^n - 1} = -0.06457$$



# Ejemplo errores DAC: cálculo INL

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida | INL(V) | INL(LSB) |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|--------|----------|
| 0      | -2,500                    | -2,400                     | -2,500                                  | -2,500                                                |        |          |
| 1      | -1,875                    | -1,700                     | -1,800                                  | -1,735                                                |        |          |
| 2      | -1,250                    | -1,000                     | -1,100                                  | -0,971                                                |        |          |
| 3      | -0,625                    | 0,200                      | 0,100                                   | 0,294                                                 |        |          |
| 4      | 0,000                     | -0,120                     | -0,220                                  | 0,038                                                 |        |          |
| 5      | 0,625                     | 1,000                      | 0,900                                   | 1,223                                                 |        |          |
| 6      | 1,250                     | 0,955                      | 0,855                                   | 1,242                                                 |        |          |
| 7      | 1,875                     | 1,523                      | 1,423                                   | 1,875                                                 |        |          |

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  | -0,452 | -0,723 | -0,06457 |
| INL             |        |        |          |

$$INL(V) = V_{off\ y\ gain_{corr}} - V_{ideal}$$

$$INL(LSB) = \frac{INL(V)}{\frac{V_{FSR}}{2^n}}$$

# Ejemplo errores DAC: cálculo INL

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida | INL(V) | INL(LSB) |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|--------|----------|
| 0      | -2,500                    | -2,400                     | -2,500                                  | -2,500                                                | 0,000  | 0,000    |
| 1      | -1,875                    | -1,700                     | -1,800                                  | -1,735                                                | 0,140  | 0,223    |
| 2      | -1,250                    | -1,000                     | -1,100                                  | -0,971                                                | 0,279  | 0,447    |
| 3      | -0,625                    | 0,200                      | 0,100                                   | 0,294                                                 | 0,919  | 1,470    |
| 4      | 0,000                     | -0,120                     | -0,220                                  | 0,038                                                 | 0,038  | 0,061    |
| 5      | 0,625                     | 1,000                      | 0,900                                   | 1,223                                                 | 0,598  | 0,957    |
| 6      | 1,250                     | 0,955                      | 0,855                                   | 1,242                                                 | -0,008 | -0,012   |
| 7      | 1,875                     | 1,523                      | 1,423                                   | 1,875                                                 | 0,000  | 0,000    |

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  | -0,452 | -0,723 | -0,06457 |
| INL             | 0,919  | 1,470  |          |

$$INL(V) = V_{off\ y\ gain_{corr}} - V_{ideal}$$

$$INL(LSB) = \frac{INL(V)}{\frac{V_{FSR}}{2^n}}$$

# Ejemplo errores DAC: cálculo DNL

| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida | INL(V) | INL(LSB) | DNL(V) | DNL(LSB) |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|--------|----------|--------|----------|
| 0      | -2,500                    | -2,400                     | -2,500                                  | -2,500                                                | 0,000  | 0,000    |        |          |
| 1      | -1,875                    | -1,700                     | -1,800                                  | -1,735                                                | 0,140  | 0,223    |        |          |
| 2      | -1,250                    | -1,000                     | -1,100                                  | -0,971                                                | 0,279  | 0,447    |        |          |
| 3      | -0,625                    | 0,200                      | 0,100                                   | 0,294                                                 | 0,919  | 1,470    |        |          |
| 4      | 0,000                     | -0,120                     | -0,220                                  | 0,038                                                 | 0,038  | 0,061    |        |          |
| 5      | 0,625                     | 1,000                      | 0,900                                   | 1,223                                                 | 0,598  | 0,957    |        |          |
| 6      | 1,250                     | 0,955                      | 0,855                                   | 1,242                                                 | -0,008 | -0,012   |        |          |
| 7      | 1,875                     | 1,523                      | 1,423                                   | 1,875                                                 | 0,000  | 0,000    |        |          |

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  | -0,452 | -0,723 | -0,06457 |
| INL             | 0,919  | 1,470  |          |
| DNL             |        |        |          |

$$DNL(V) = (V_{o \text{ y } g_{cor}(CD)} - V_{o \text{ y } g_{cor}(CD-1)}) - \frac{V_{FSR}}{2^n}$$

$$DNL(LSB) = \frac{DNL(V)}{\frac{V_{FSR}}{2^n}}$$

# Ejemplo errores DAC: cálculo DNL

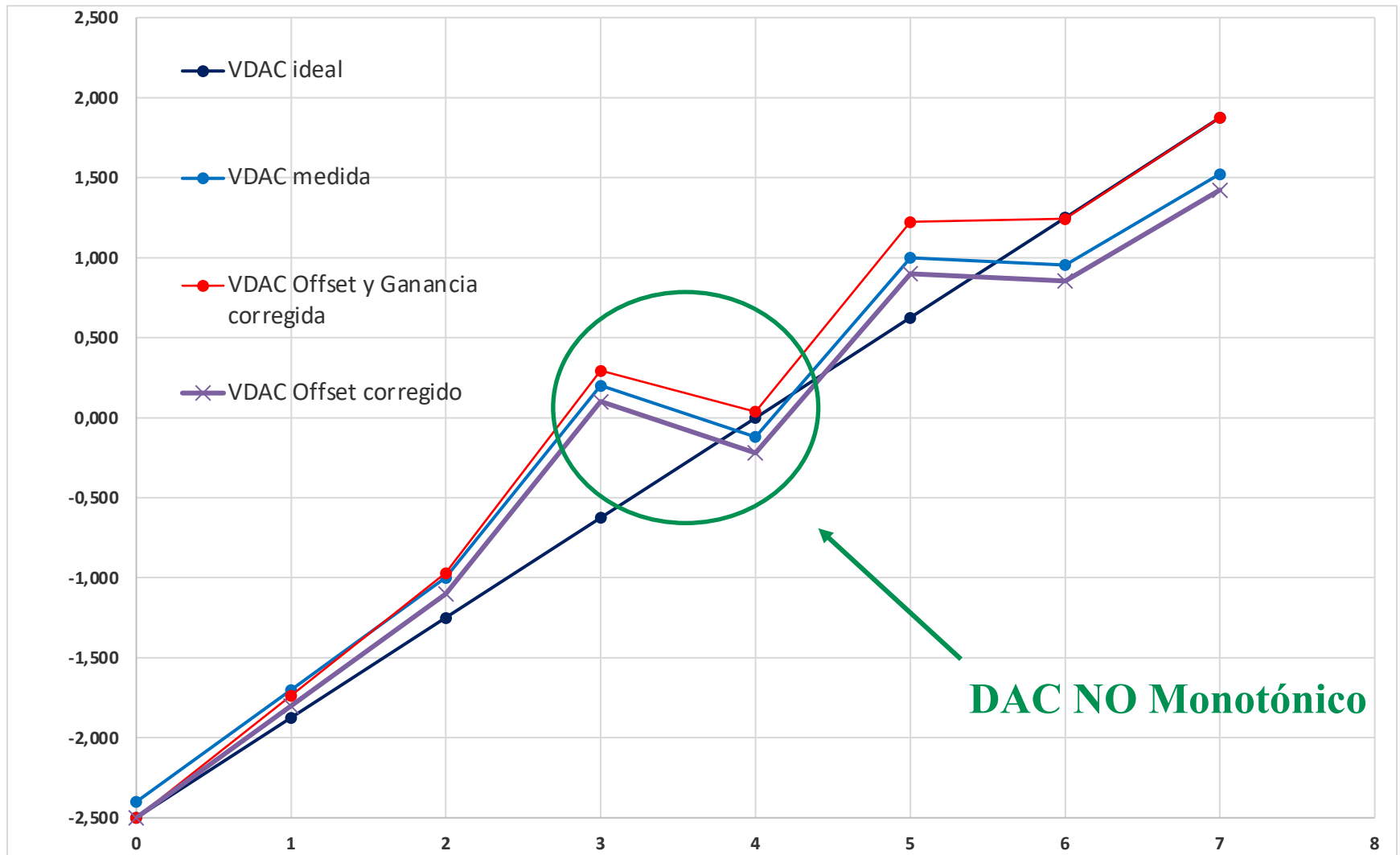
| Código | V <sub>DAC</sub><br>Ideal | V <sub>DAC</sub><br>Medida | V <sub>DAC</sub><br>Offset<br>corregido | V <sub>DAC</sub><br>Offset y<br>Ganancia<br>corregida | INL(V) | INL(LSB) | DNL(V) | DNL(LSB) |
|--------|---------------------------|----------------------------|-----------------------------------------|-------------------------------------------------------|--------|----------|--------|----------|
| 0      | -2,500                    | -2,400                     | -2,500                                  | -2,500                                                | 0,000  | 0,000    |        |          |
| 1      | -1,875                    | -1,700                     | -1,800                                  | -1,735                                                | 0,140  | 0,223    | 0,140  | 0,223    |
| 2      | -1,250                    | -1,000                     | -1,100                                  | -0,971                                                | 0,279  | 0,447    | 0,140  | 0,223    |
| 3      | -0,625                    | 0,200                      | 0,100                                   | 0,294                                                 | 0,919  | 1,470    | 0,640  | 1,023    |
| 4      | 0,000                     | -0,120                     | -0,220                                  | 0,038                                                 | 0,038  | 0,061    | -0,880 | -1,409   |
| 5      | 0,625                     | 1,000                      | 0,900                                   | 1,223                                                 | 0,598  | 0,957    | 0,560  | 0,895    |
| 6      | 1,250                     | 0,955                      | 0,855                                   | 1,242                                                 | -0,008 | -0,012   | -0,605 | -0,969   |
| 7      | 1,875                     | 1,523                      | 1,423                                   | 1,875                                                 | 0,000  | 0,000    | 0,008  | 0,012    |

|                 | V      | LSB    | V/código |
|-----------------|--------|--------|----------|
| Error de offset | 0,100  | 0,160  |          |
| Error de cero   | -0,120 | -0,192 |          |
| Error Ganancia  | -0,452 | -0,723 | -0,06457 |
| INL             | 0,919  | 1,470  |          |
| DNL             | 0,880  | 1,409  |          |

$$DNL(V) = (V_{o y g_{cor}(CD)} - V_{o y g_{cor}(CD-1)}) - \frac{V_{FSR}}{2^n}$$

$$DNL(LSB) = \frac{DNL(V)}{\frac{V_{FSR}}{2^n}}$$

# Ejemplo errores DAC



### FEATURES

**Micropower:** 100  $\mu$ A  
**0.1  $\mu$ A typical power shutdown**  
**Single-supply 2.7 V to 5.5 V operation**  
**AD7392:** 12-bit resolution  
**AD7393:** 10-bit resolution  
**0.9 LSB differential nonlinearity error**

### APPLICATIONS

**Automotive 0.5 V to 4.5 V output span voltage**  
**Portable communications**  
**Digitally controlled calibration**  
**PC peripherals**

### FUNCTIONAL BLOCK DIAGRAM

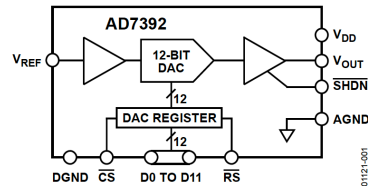


Figure 1.

## SPECIFICATIONS

### ELECTRICAL CHARACTERISTICS

At  $V_{REF} = 2.5$  V,  $-40^{\circ}\text{C} < T_A < +85^{\circ}\text{C}$ , unless otherwise noted.

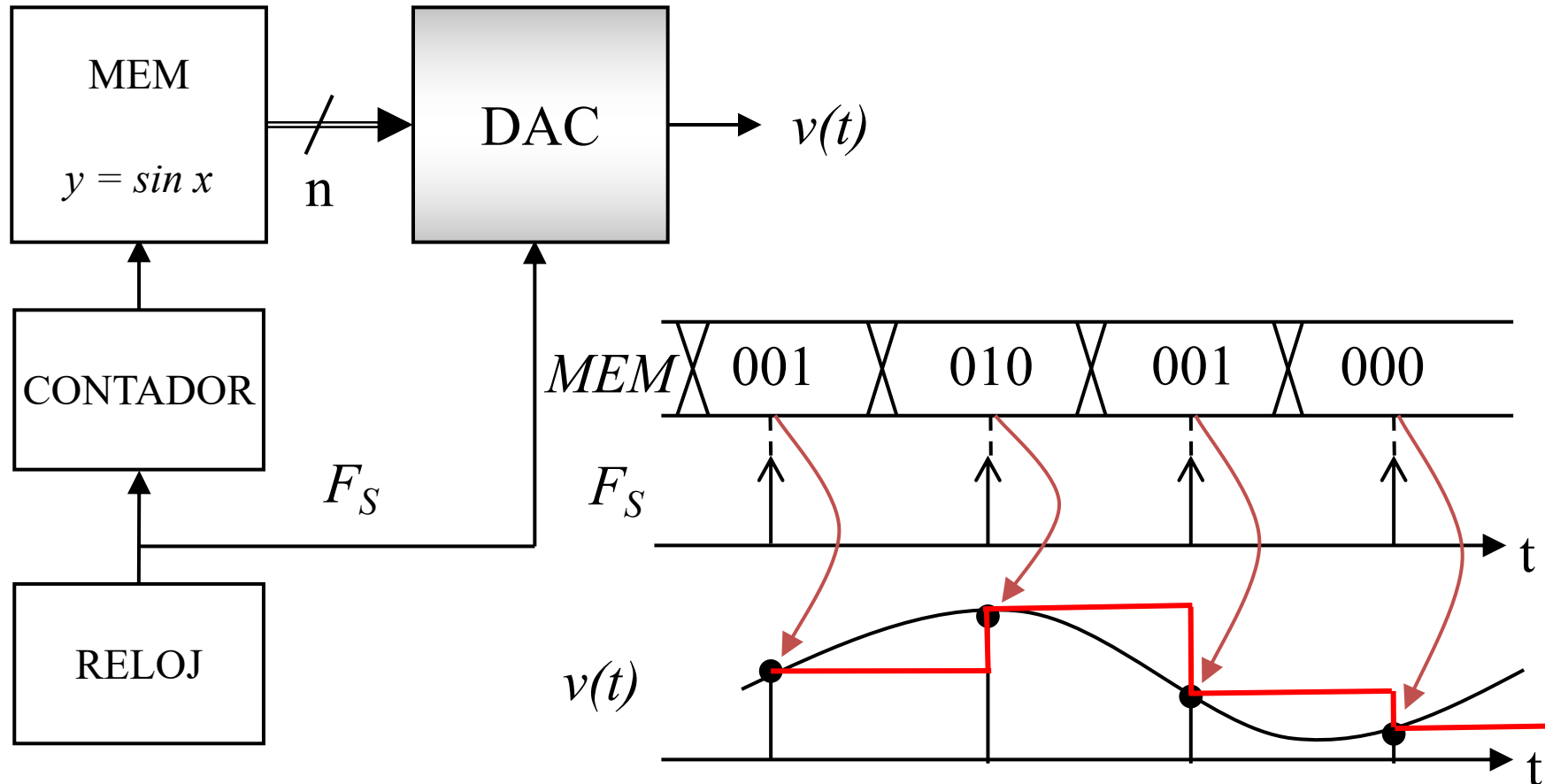
Table 1. AD7392

| Parameter                                       | Symbol     | Conditions                                                       | 3 V $\pm$ 10% | 5 V $\pm$ 10% | Unit                        |
|-------------------------------------------------|------------|------------------------------------------------------------------|---------------|---------------|-----------------------------|
| STATIC PERFORMANCE                              |            |                                                                  |               |               |                             |
| Resolution <sup>1</sup>                         | N          |                                                                  | 12            | 12            | Bits                        |
| Relative Accuracy <sup>2</sup>                  | INL        | $T_A = +25^{\circ}\text{C}$                                      | $\pm 1.8$     | $\pm 1.8$     | LSB max                     |
|                                                 |            | $T_A = -40^{\circ}\text{C}, +85^{\circ}\text{C}$                 | $\pm 3$       | $\pm 3$       | LSB max                     |
| Differential Nonlinearity <sup>2</sup>          | DNL        | $T_A = +25^{\circ}\text{C}$ , monotonic                          | $\pm 0.9$     | $\pm 0.9$     | LSB max                     |
|                                                 |            | Monotonic                                                        | $\pm 1$       | $\pm 1$       | LSB max                     |
| Zero-Scale Error                                | $V_{ZSE}$  | Data = 0x000, $T_A = +25^{\circ}\text{C}, +85^{\circ}\text{C}$   | 4.0           | 4.0           | mV max                      |
|                                                 |            | Data = 0x000, $T_A = -40^{\circ}\text{C}$                        | 8.0           | 8.0           | mV max                      |
| Full-Scale Voltage Error                        | $V_{FSE}$  | $T_A = +25^{\circ}\text{C}, +85^{\circ}\text{C}$ , data = 0xFFFF | $\pm 8$       | $\pm 8$       | mV max                      |
|                                                 |            | $T_A = -40^{\circ}\text{C}$ , data = 0xFFFF                      | $\pm 20$      | $\pm 20$      | mV max                      |
| Full-Scale Temperature Coefficient <sup>3</sup> | $TCV_{FS}$ |                                                                  | 28            | 28            | ppm/ $^{\circ}\text{C}$ typ |

# Índice

1. Introducción
2. Especificaciones estáticas
- 3. Especificaciones dinámicas**
4. Técnicas de conversión
5. Ejemplos: DAC08 y TLC5620C

# Ejemplo: DDS (Direct Digital Synthesizer)



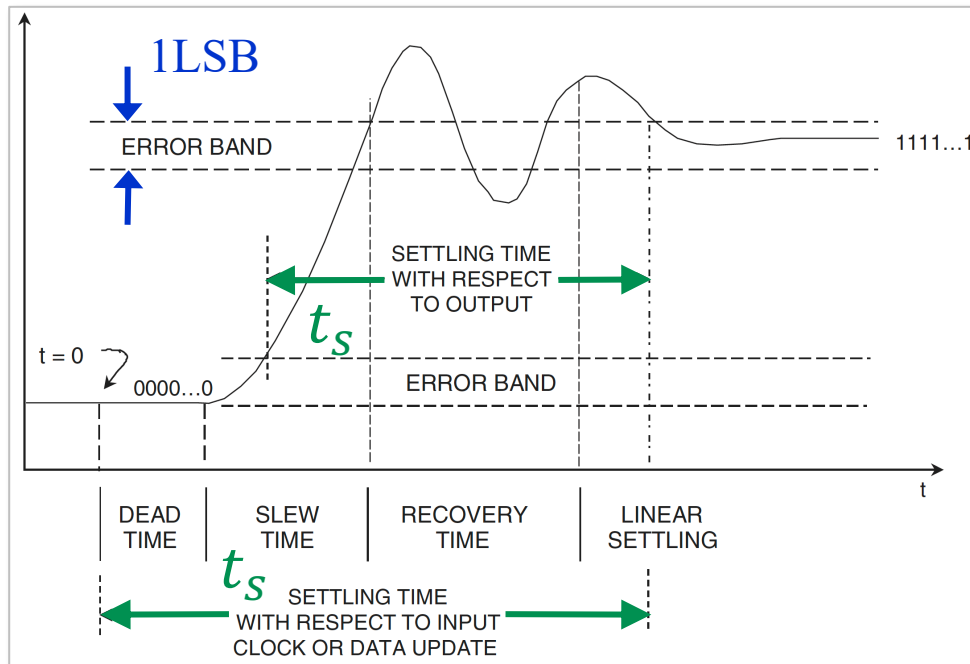
Pero...

- ¿Cada cuánto tiempo se puede generar una nueva muestra?
- La forma de onda de salida, ¿realmente es así?



# Tiempo para generar una nueva muestra:

## Tiempo de establecimiento ( $t_s$ )



## SPECIFICATIONS

### ELECTRICAL CHARACTERISTICS

At  $V_{REF} = 2.5\text{ V}$ ,  $-40^\circ\text{C} < T_A < +85^\circ\text{C}$ , unless otherwise noted.

Table 1. AD7392

| Parameter                  | Symbol    | Conditions                      | 3 V $\pm 10\%$ | 5 V $\pm 10\%$ | Unit                 |
|----------------------------|-----------|---------------------------------|----------------|----------------|----------------------|
| AC CHARACTERISTICS         |           |                                 |                |                |                      |
| Output Slew Rate           | SR        | Data = 0x000 to 0xFFFF to 0x000 | 0.05           | 0.05           | V/ $\mu\text{s}$ typ |
| Settling Time <sup>6</sup> | $t_s$     | To $\pm 0.1\%$ of full scale    | 70             | 60             | $\mu\text{s}$ typ    |
| Shutdown Recovery Time     | $t_{SDR}$ |                                 |                | 80             | $\mu\text{s}$ typ    |



3 V, Parallel Input  
Micropower 10-/12-Bit DACs

AD7392/AD7393

#### FEATURES

Micropower: 100  $\mu\text{A}$   
0.1  $\mu\text{A}$  typical power shutdown  
Single-supply 2.7 V to 5.5 V operation  
AD7392: 12-bit resolution  
AD7393: 10-bit resolution  
0.9 LSB differential nonlinearity error

#### APPLICATIONS

Automotive 0.5 V to 4.5 V output span voltage  
Portable communications  
Digitally controlled calibration  
PC peripherals

#### FUNCTIONAL BLOCK DIAGRAM

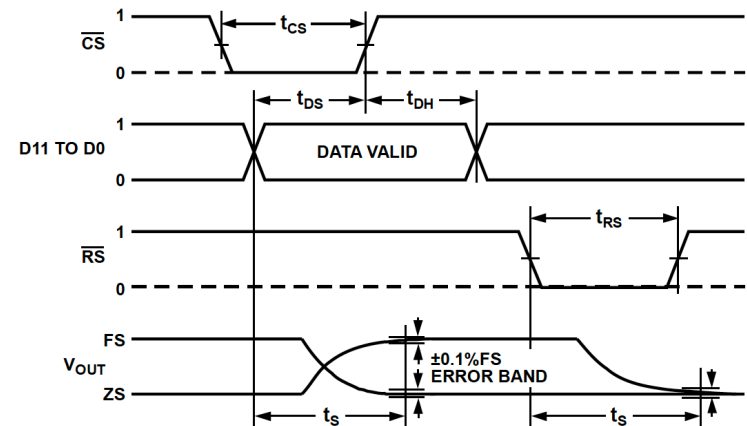
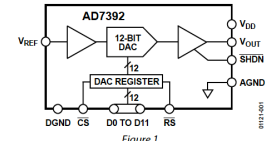
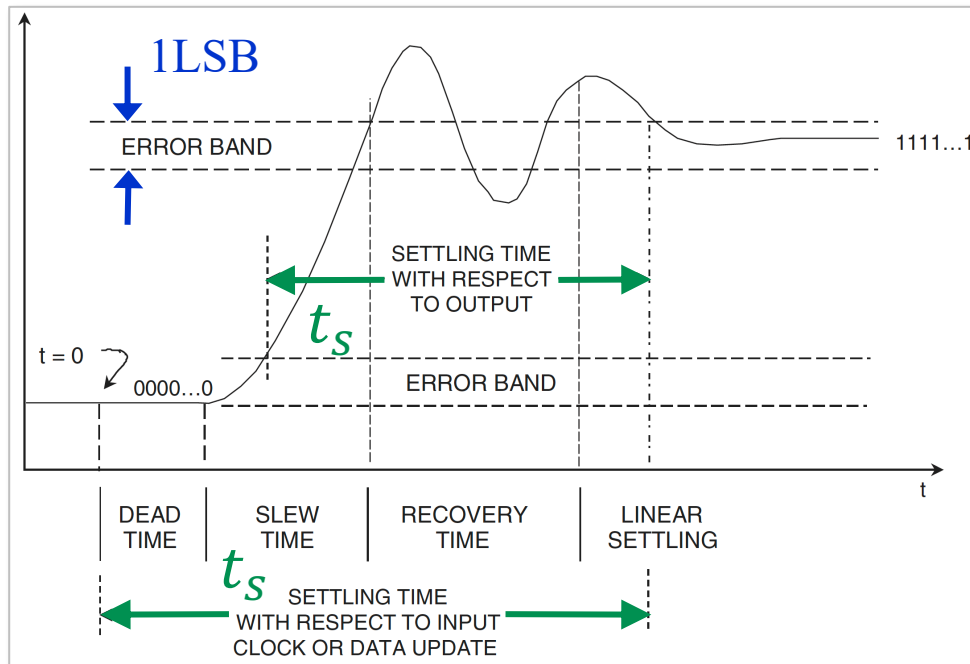


Figure 2. Timing Diagram

## Ejercicio: Tiempo de establecimiento ( $t_s$ )



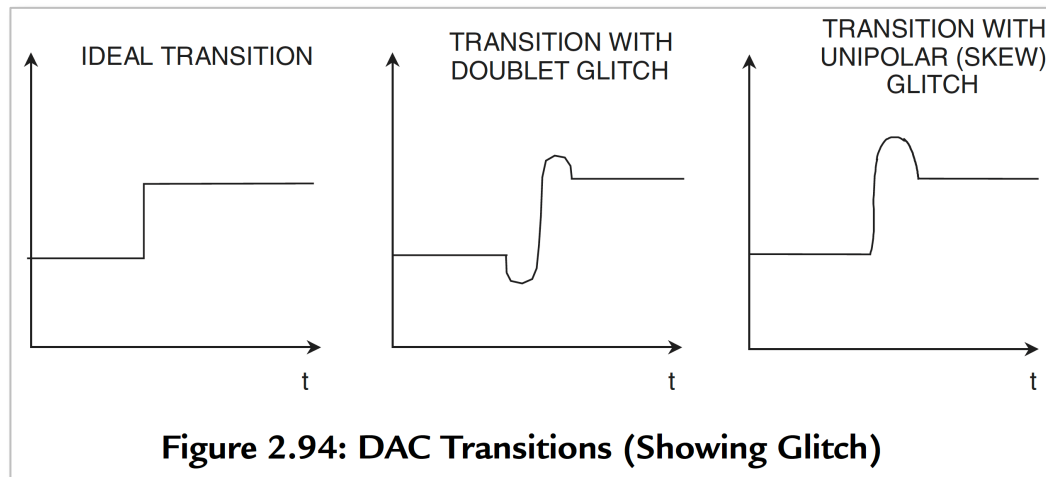
Pensemos en un display digital FHD (1920x1080 pixel) que trabaja con un refresco de 60 Hz.

¿Qué valor máximo de  $t_s$  deberá tener el DAC que active cada uno de los pixeles?

**Dato:** utilice, en su cálculo, una holgura del 35 %.

## Forma de onda entre muestras: Glitch Impulse Area

Cuando cambia la salida de un DAC se puede observar en ella un **Glitch** debido a un sobreimpulso, un subimpulso o ambos.

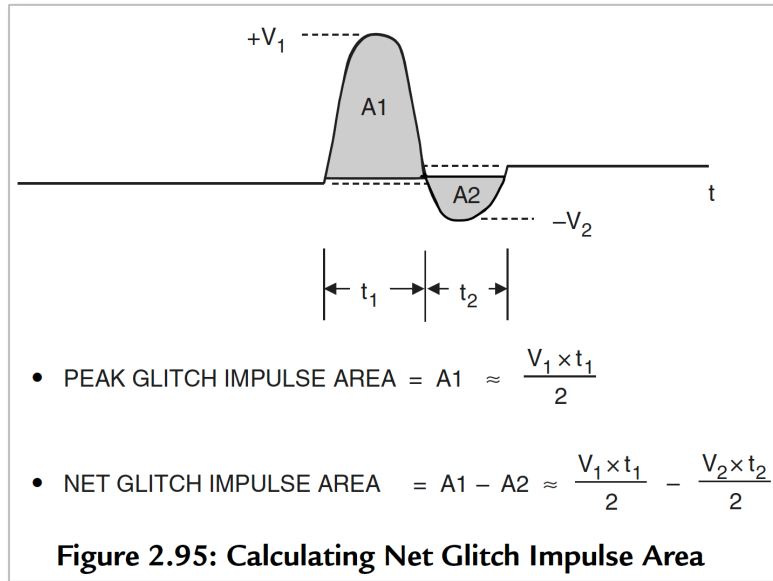


¿A qué son debidos?:

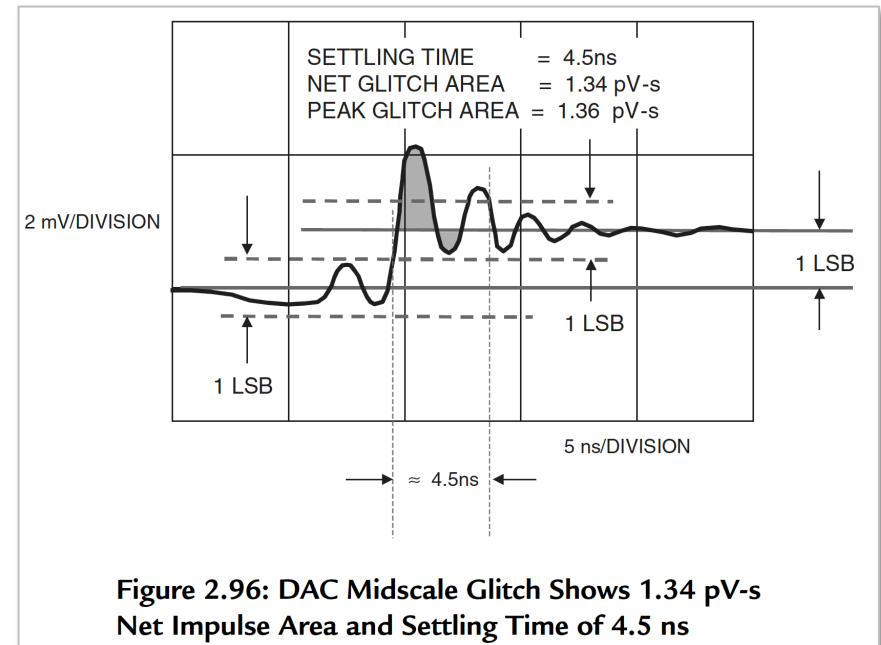
- Por acoplo capacitivo en la salida desde las transiciones digitales. Generalmente positivos y negativos que en promedio se cancelan. Menos importantes.
- Spúreos en la salida debido a la diferente velocidad de los interruptores internos del DAC. Generalmente unipolares, más grandes y más importantes.

# Forma de onda entre muestras: Glitch Impulse Area

Para caracterizar estos Glitch se mide su área: **Glitch Impluse Area**.



En el medio del FSR se produce el mayor Glitch:  
011...111 a 100...000



3 V, Parallel Input  
Micropower 10-/12-Bit DACs  
AD7392/AD7393

## SPECIFICATIONS

### ELECTRICAL CHARACTERISTICS

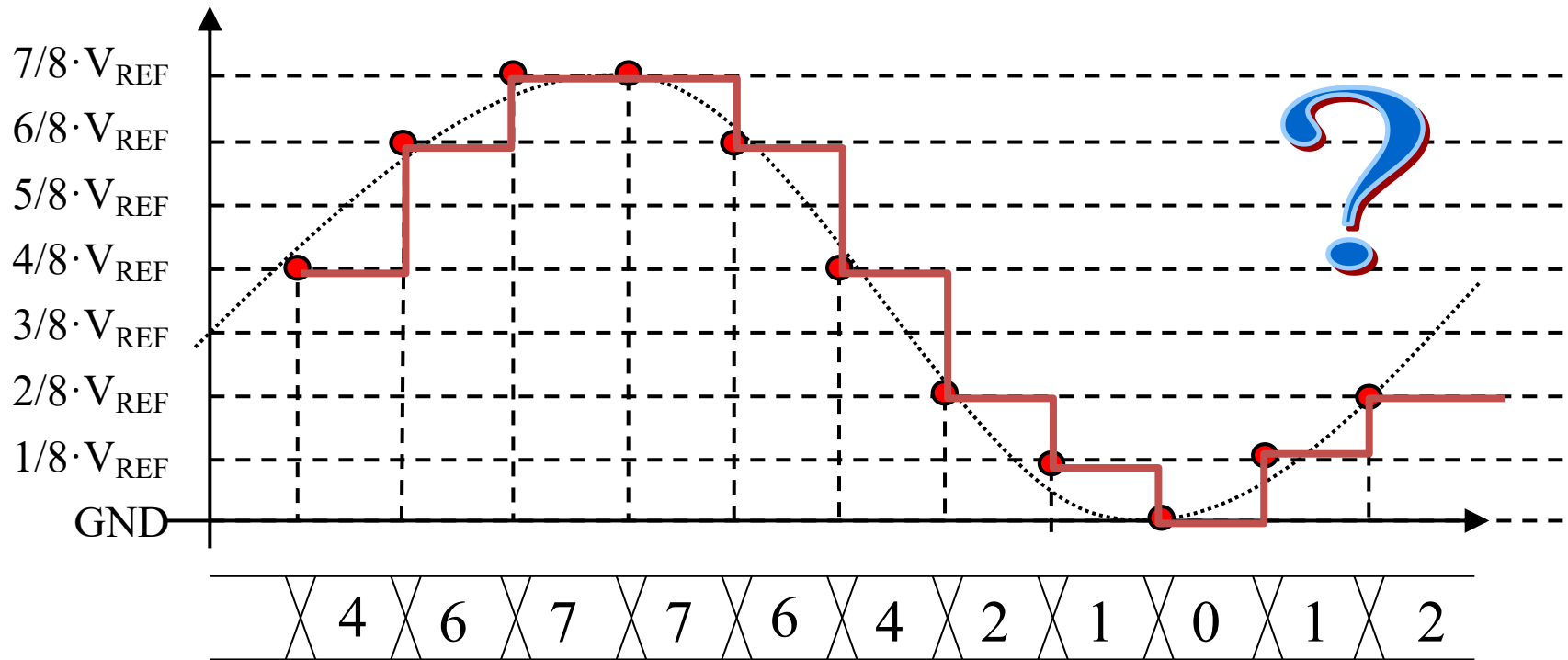
At  $V_{REF} = 2.5\text{ V}$ ,  $-40^\circ\text{C} < T_A < +85^\circ\text{C}$ , unless otherwise noted.

Table 1. AD7392

| Parameter          | Symbol | Conditions                             | 3 V $\pm$ 10% | 5 V $\pm$ 10% | Unit     |
|--------------------|--------|----------------------------------------|---------------|---------------|----------|
| AC CHARACTERISTICS |        |                                        |               |               |          |
| DAC Glitch         |        | Code 0x7FF to Code 0x800 to Code 0x7FF | 65            | 65            | nV/s typ |

# ¿Cómo será la forma de onda a la salida del DAC (del DDS)?

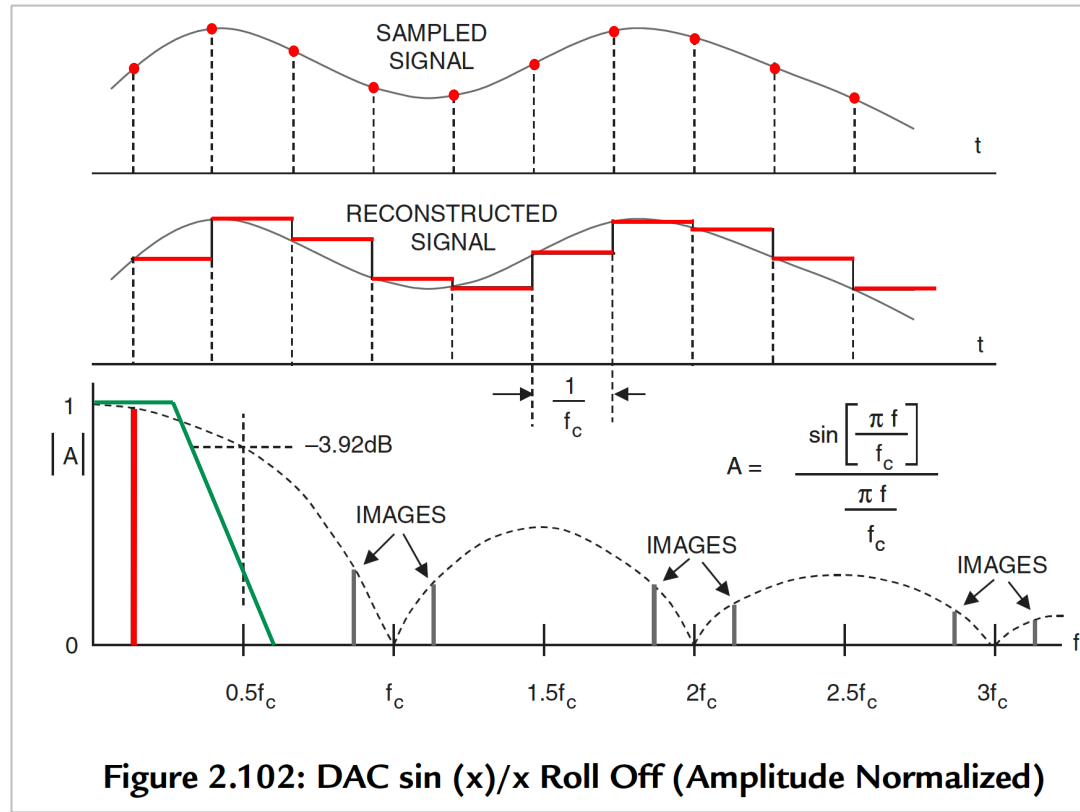
Como la salida del convertidor DAC sólo puede tomar un conjunto finito de valores, su aspecto en el tiempo será:



Convertidor D/A de 3 bits con salida unipolar,  $k=1$

# ¿Cómo será la forma de onda a la salida del DAC (del DDS)?

Visto en el dominio de la frecuencia:



Se necesita, a la salida del DAC, un filtro de reconstrucción: un filtro de paso bajo.

# ¿Cómo será la forma de onda a la salida del DAC (del DDS)?

Pero..., la señal reconstruida a la salida del DDS, además irá acompañada de:

**Ruido** y **Productos de distorsión**.

La **distorsión** se puede especificar en términos de:

- Distorsión armónica (THD).
- Spurious Free Dynamic Range (SFDR).
- Distorsión de intermodulación.

¿Qué errores contribuyen a la **distorsión**?

- La resolución.
- El INL y el DNL.
- Glitches dependientes del código.
- La relación entre la frecuencia del reloj ( $f_s$ ) y la frecuencia de salida ( $f_c$ ).
- ...

# ¿Cómo será la forma de onda a la salida del DAC (del DDS)?

¿Cómo se mide el ruido y la distorsión?

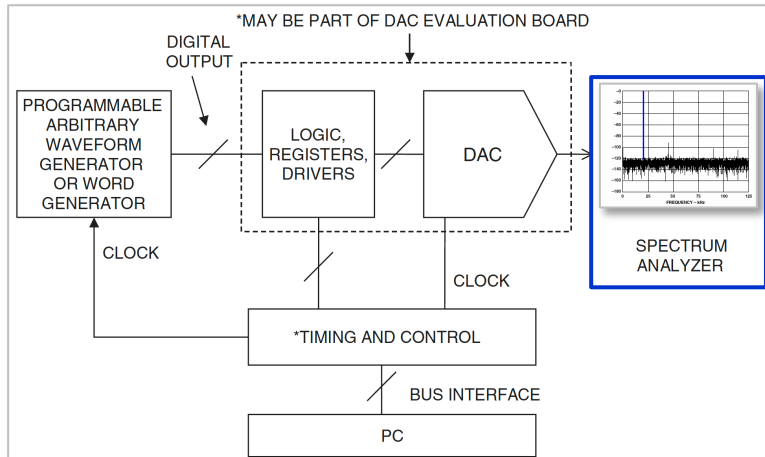


Figure 5.16: Test Setup for Measuring DAC Distortion and Noise

## Parámetros de interés

- $SNR = S/NOISE\ FLOOR - 10 \log_{10} \left[ \frac{f_s/2}{BW} \right]$
- $THD = 20 \log_{10} \sqrt{10^{-V2/20}^2 + 10^{-V3/20}^2 + \dots + 10^{-V6/20}^2}$
- $SINAD = 20 \log_{10} \sqrt{10^{-SNR/20}^2 + 10^{-THD/20}^2}$

NOTE: NOISE FLOOR, SNR, THD, SINAD, V2, V3, ... , V6 in units of dBc

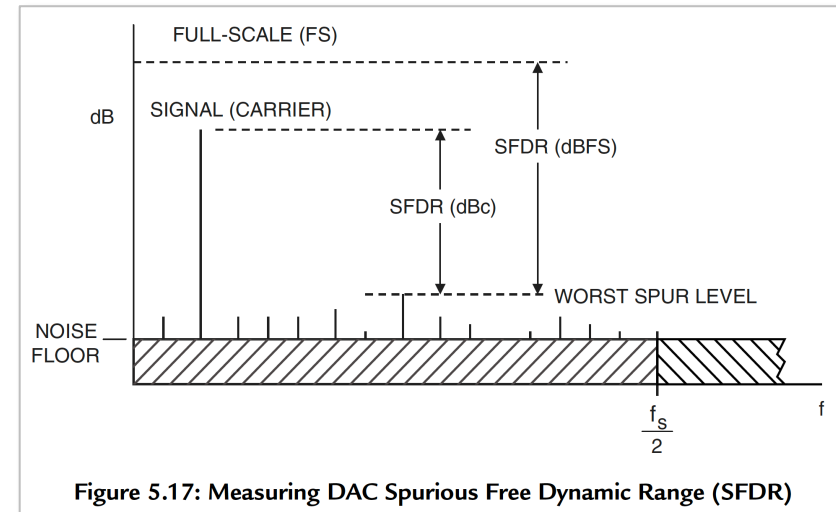


Figure 5.17: Measuring DAC Spurious Free Dynamic Range (SFDR)

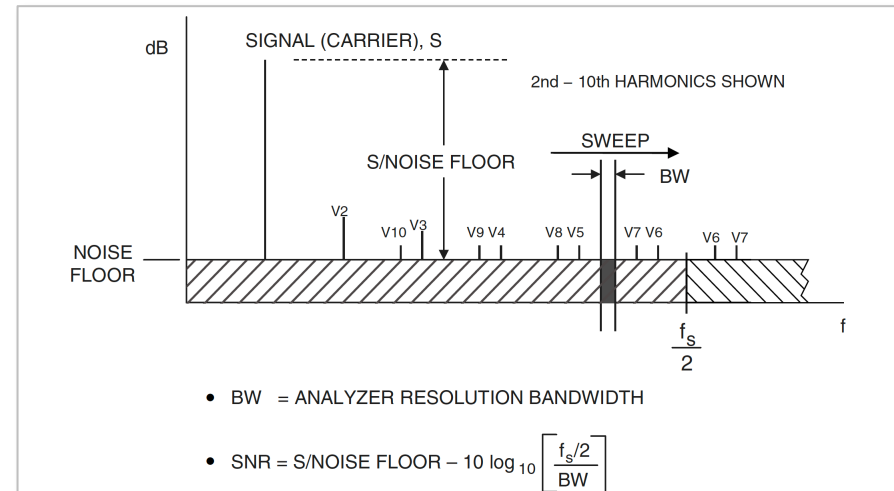


Figure 5.18: Measuring DAC Distortion and SNR with an Analog Spectrum Analyzer





FEATURES

- High performance member of pin-compatible TxDAC product family
- Excellent spurious-free dynamic range performance SFDR to Nyquist
  - 83 dBc at 5 MHz output
  - 80 dBc at 10 MHz output
  - 73 dBc at 20 MHz output
- SNR at 5 MHz output, 125 MSPS: 77 dB
- Twos complement or straight binary data format
- Differential current outputs: 2 mA to 20 mA
- Power dissipation: 135 mW at 3.3 V
- Power-down mode: 15 mW at 3.3 V
- On-chip 1.2 V reference
- CMOS-compatible digital interface
- 28-lead SOIC, 28-lead TSSOP, and 32-lead LFCSP packages
- Edge-triggered latches

APPLICATIONS

- Wideband communication transmit channel
  - Direct IFs
  - Base stations
  - Wireless local loops
  - Digital radio links
  - Direct digital synthesis (DDS)
  - Instrumentation

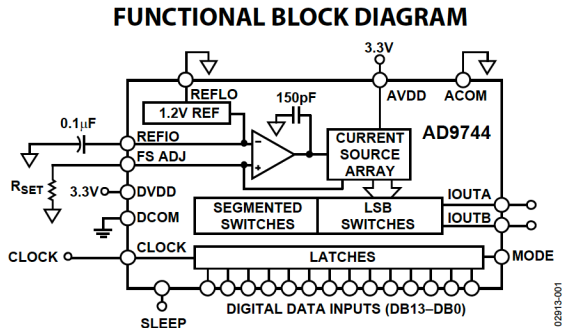


Figure 1.

DYNAMIC SPECIFICATIONS

T<sub>MIN</sub> to T<sub>MAX</sub>, AVDD = 3.3 V, DVDD = 3.3 V, CLKVDD = 3.3 V, I<sub>OUTFS</sub> = 20 mA, differential transformer coupled output, 50 Ω doubly terminated, unless otherwise noted.

Table 2.

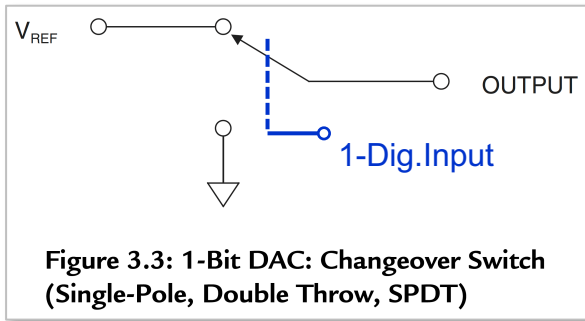
| Parameter                                                                           | Min | Typ | Max | Unit |
|-------------------------------------------------------------------------------------|-----|-----|-----|------|
| AC LINEARITY                                                                        |     |     |     |      |
| Spurious-Free Dynamic Range to Nyquist                                              |     |     |     |      |
| f <sub>CLOCK</sub> = 25 MSPS; f <sub>OUT</sub> = 1.00 MHz                           | 77  | 90  |     | dBc  |
| 0 dBFS Output                                                                       |     |     |     |      |
| −6 dBFS Output                                                                      |     |     |     |      |
| −12 dBFS Output                                                                     |     |     |     |      |
| −18 dBFS Output                                                                     |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 1.00 MHz                           |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 2.51 MHz                           |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 10 MHz                             |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 15 MHz                             |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 25 MHz                             |     |     |     |      |
| f <sub>CLOCK</sub> = 165 MSPS; f <sub>OUT</sub> = 21 MHz                            | 85  |     |     | dBc  |
| f <sub>CLOCK</sub> = 165 MSPS; f <sub>OUT</sub> = 41 MHz                            | 84  |     |     | dBc  |
| f <sub>CLOCK</sub> = 210 MSPS; f <sub>OUT</sub> = 41 MHz                            | 80  |     |     | dBc  |
| f <sub>CLOCK</sub> = 210 MSPS; f <sub>OUT</sub> = 69 MHz                            | 75  |     |     | dBc  |
| Spurious-Free Dynamic Range Within a Window                                         |     |     |     |      |
| f <sub>CLOCK</sub> = 25 MSPS; f <sub>OUT</sub> = 1.00 MHz; 2 MHz Span               | 84  | 90  |     | dBc  |
| f <sub>CLOCK</sub> = 50 MSPS; f <sub>OUT</sub> = 5.02 MHz; 2 MHz Span               |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 5.03 MHz; 2.5 MHz Span             |     |     |     |      |
| f <sub>CLOCK</sub> = 125 MSPS; f <sub>OUT</sub> = 5.04 MHz; 4 MHz Span              |     |     |     |      |
| Total Harmonic Distortion                                                           |     |     |     |      |
| f <sub>CLOCK</sub> = 25 MSPS; f <sub>OUT</sub> = 1.00 MHz                           | −77 | −86 |     | dBc  |
| f <sub>CLOCK</sub> = 50 MSPS; f <sub>OUT</sub> = 2.00 MHz                           |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 2.00 MHz                           |     |     |     |      |
| f <sub>CLOCK</sub> = 125 MSPS; f <sub>OUT</sub> = 2.00 MHz                          |     |     |     |      |
| Signal-to-Noise Ratio                                                               |     |     |     |      |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 20 mA  | 82  | 88  |     | dB   |
| f <sub>CLOCK</sub> = 65 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 5 mA   |     |     |     |      |
| f <sub>CLOCK</sub> = 125 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 20 mA |     |     |     |      |
| f <sub>CLOCK</sub> = 125 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 5 mA  |     |     |     |      |
| f <sub>CLOCK</sub> = 165 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 20 mA |     |     |     |      |
| f <sub>CLOCK</sub> = 165 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 5 mA  |     |     |     |      |
| f <sub>CLOCK</sub> = 210 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 20 mA |     |     |     |      |
| f <sub>CLOCK</sub> = 210 MSPS; f <sub>OUT</sub> = 5 MHz; I <sub>OUTFS</sub> = 5 mA  |     |     |     |      |

# Índice

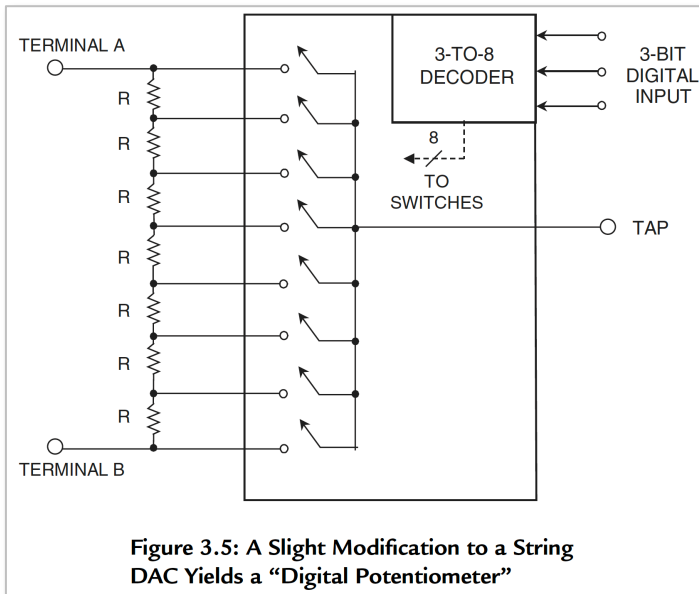
1. Introducción
2. Especificaciones estáticas
3. Especificaciones dinámicas
- 4. Técnicas de conversión**
5. Ejemplos: DAC08 y TLC5620C

# Primeras estructuras

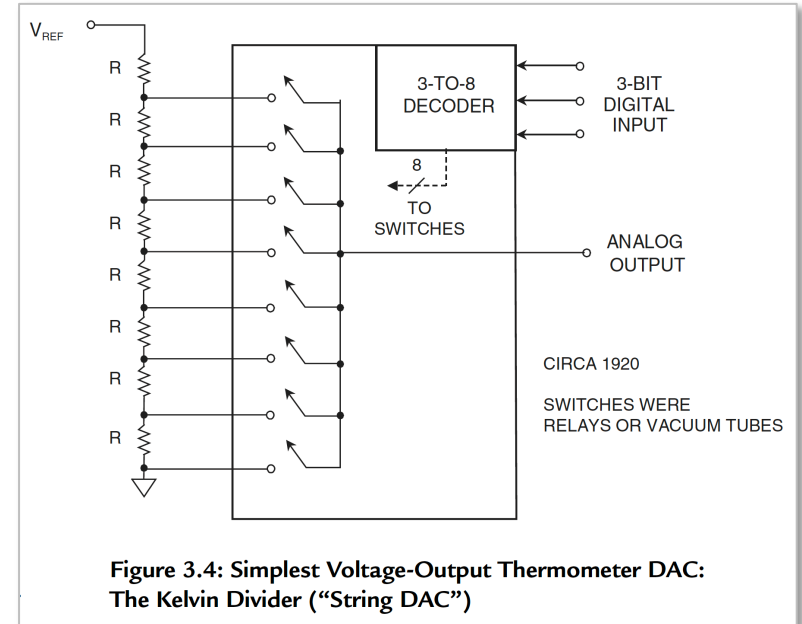
## DAC de 1-Bit.



## Potenciómetro Digital.



## DAC de 3-Bit.

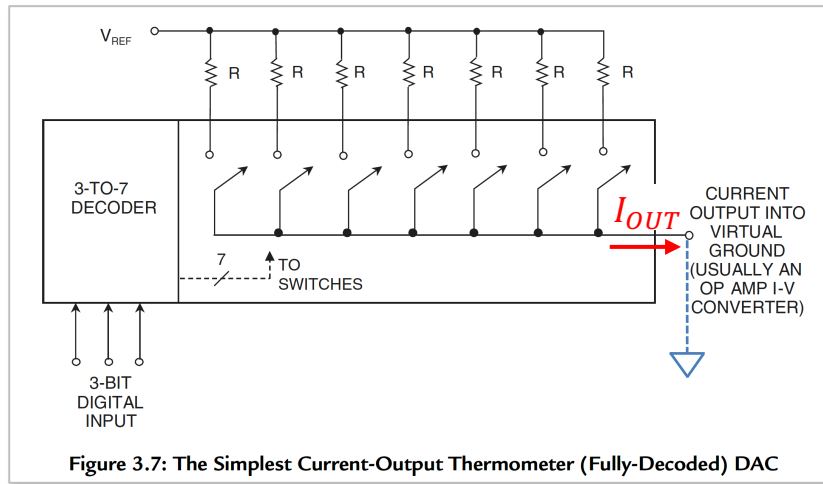


$$V_{OUT} = V_{REF} \frac{VDC}{2^n}$$

- Con  $2^3$  resistores e interruptores.
- $R_{out}$  depende del código.
- Es monótonico.
- Con  $R$ 's iguales es lineal.
- Los Glitch son bajos (sólo dos interruptores se mueven) y no dependen del código.

# Primeras estructuras

## DAC de 3-Bit. (Salida en corriente)

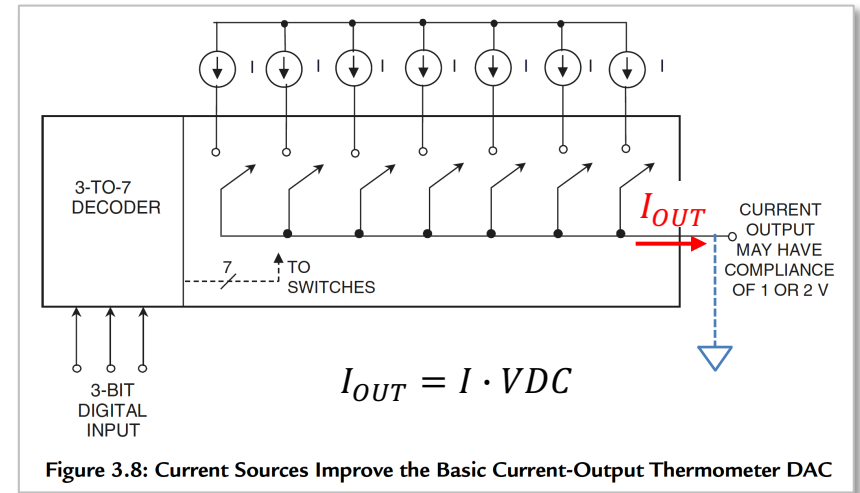


$$I_{OUT} = \frac{V_{REF}}{R} \cdot VDC$$

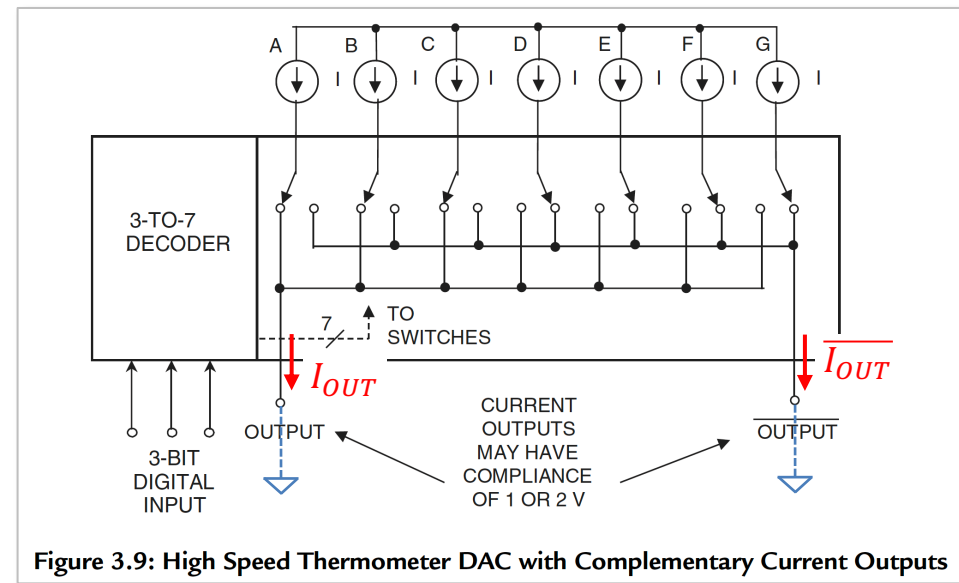
- Con  $(2^3-1)$  resistores e interruptores.
- Es monotónico.

$$I_{OUT} = I \cdot VDC$$

$$\overline{I_{OUT}} = I \cdot \overline{VDC}$$



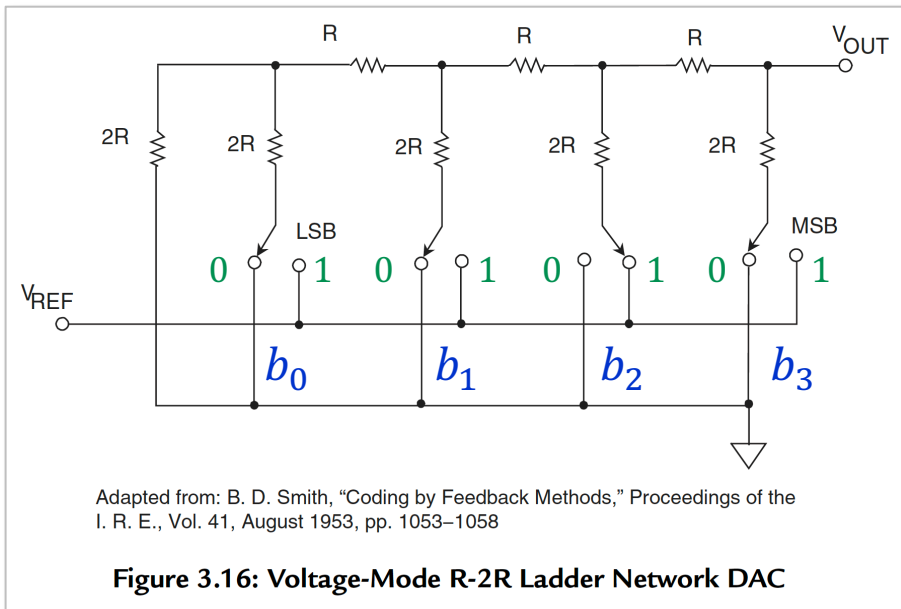
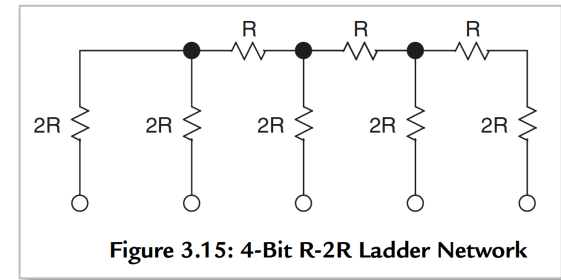
## DAC de 3-Bit. (Salida en corriente doble)



# Red en escalera R-2R

Es la estructura más habitual, empleada en la construcción de DAC.

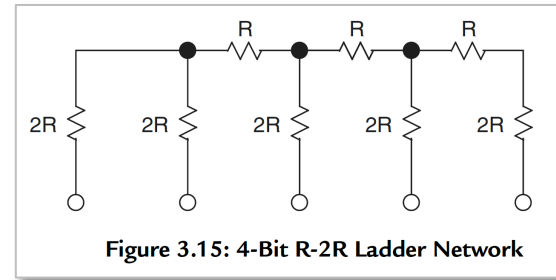
DAC en modo tensión:



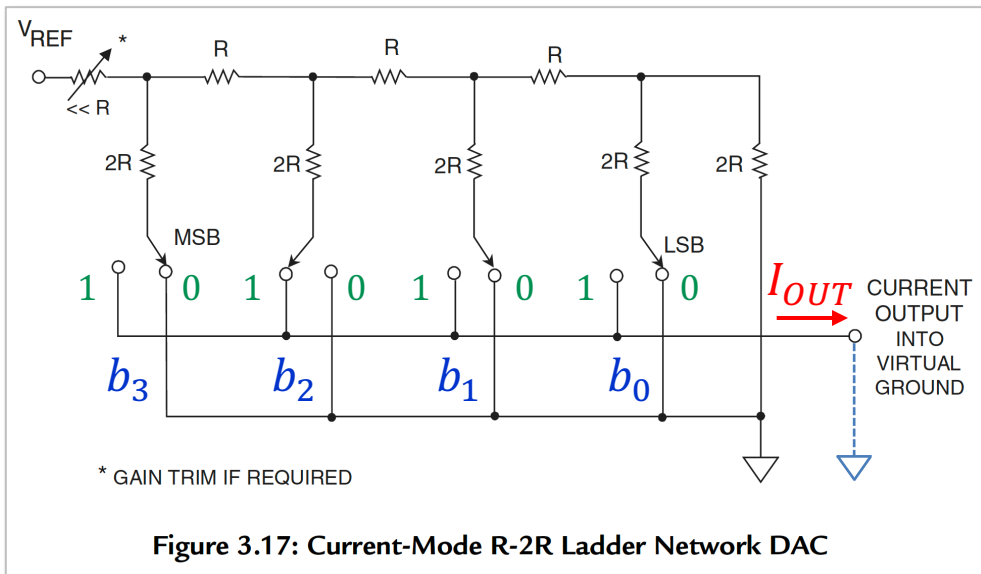
$$V_{OUT} = V_{REF} \frac{VDC}{2^n}$$

# Red en escalera R-2R

Es la estructura más habitual, empleada en la construcción de DAC.



## DAC en modo corriente:



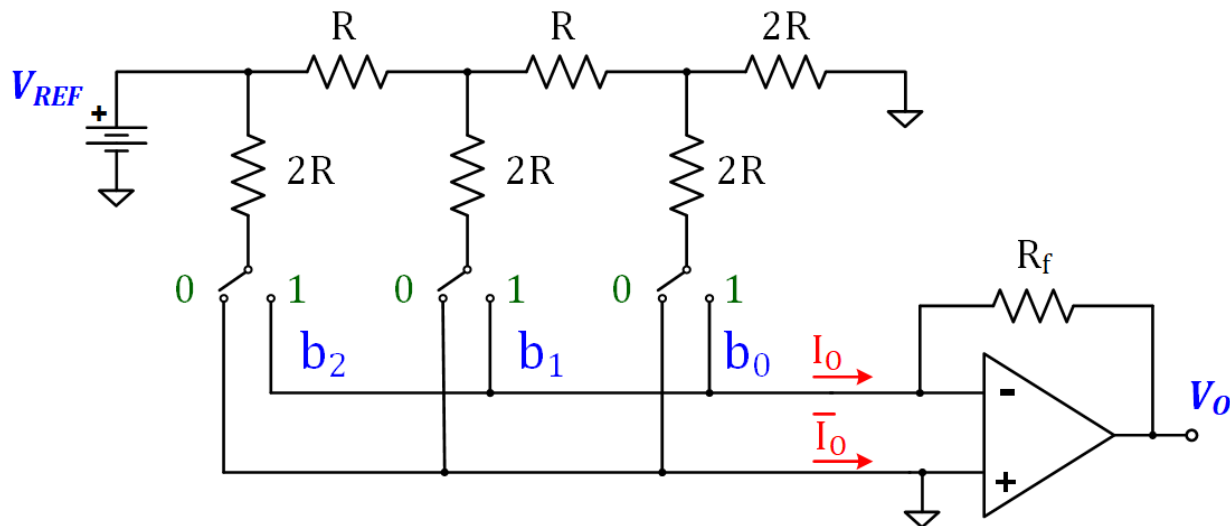
$$I_{OUT} = \frac{V_{REF}}{R} \cdot \frac{VDC}{2^n}$$

La  $R^*$  en serie con  $V_{REF}$  permite ajustar su ganancia.

$$I_{OUT} = \frac{V_{REF}}{(R + R^*)} \cdot \frac{VDC}{2^n}$$

# Red en escalera R-2R: DAC de 3 bit.

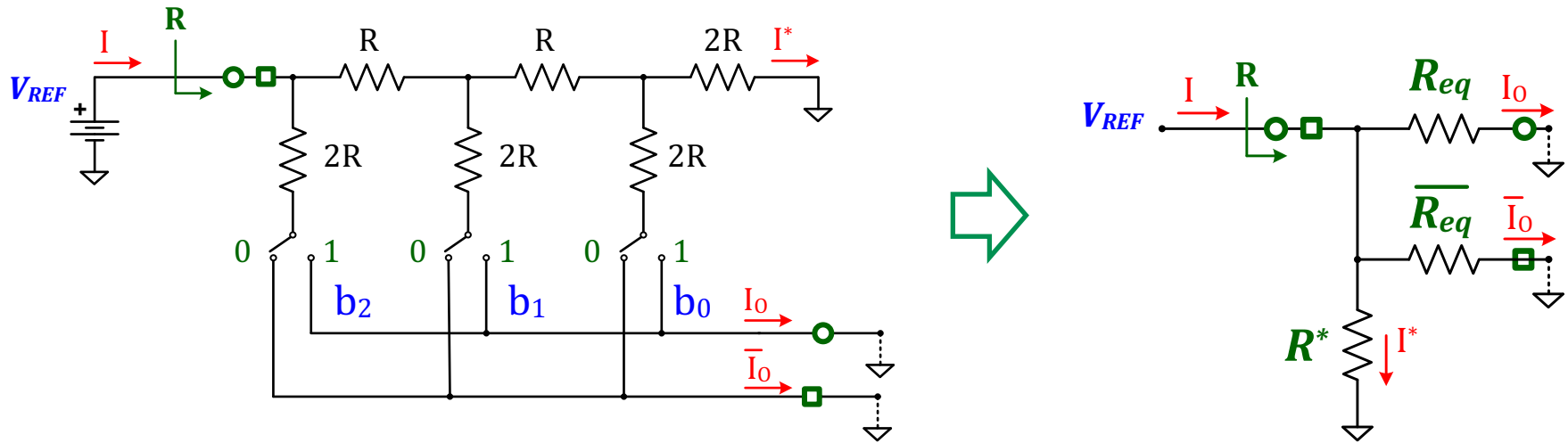
Ejemplo de utilización de una red en escalera R-2R, en modo corriente, para la construcción de un DAC de 3 bit con salida en tensión.



¿Cuál es la expresión de  $V_o$ ?

# Red en escalera R-2R: Detalle.

Observe que la red en escalera R-2R equivale a una resistencia  $R_{eq}$  y  $\overline{R_{eq}}$  entre el terminal de conexión de  $V_{REF}$  y cada salida.

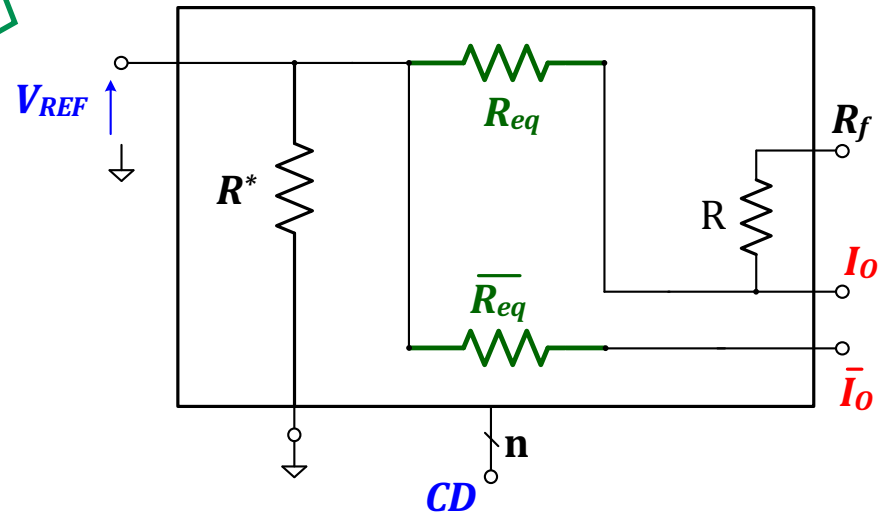
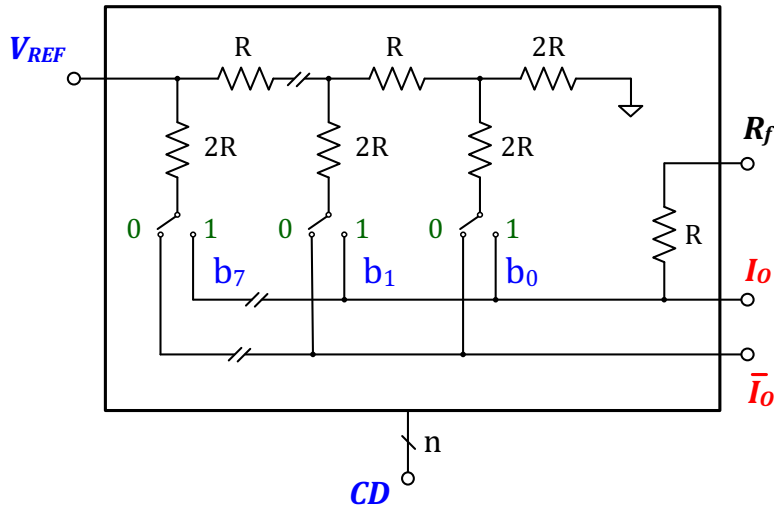


¿Cuál es la expresión de  $R_{eq}$ ,  $\overline{R_{eq}}$  y  $R^*$ ?



# Red en escalera R-2R: DAC de tipo multiplicador.

DAC de tipo multiplicador de n bits.



$$R_{eq} = R \cdot \frac{2^n}{VDC}$$

$$\overline{R_{eq}} = R \cdot \frac{2^n}{\overline{VDC}}$$

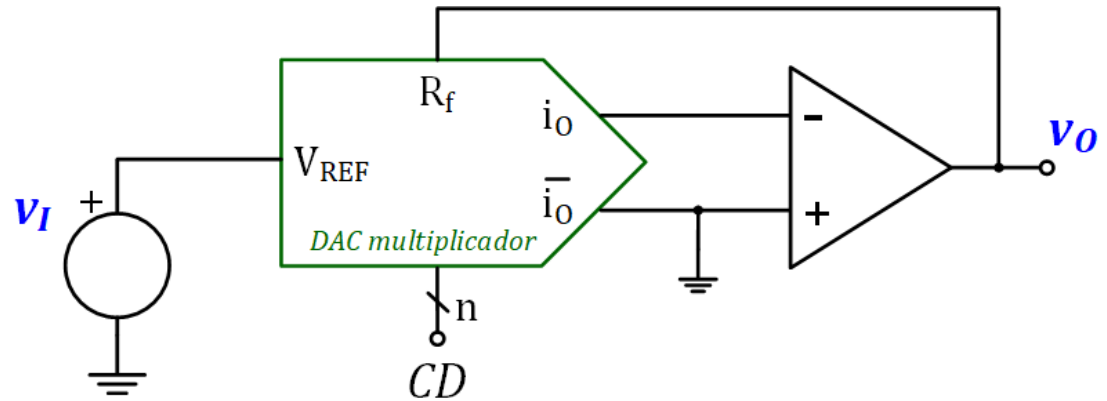
$$R^* = R \cdot 2^n$$

$VDC$ : Valor Decimal del Código

$\overline{VDC}$ : Valor Decimal del Código en complemento a uno

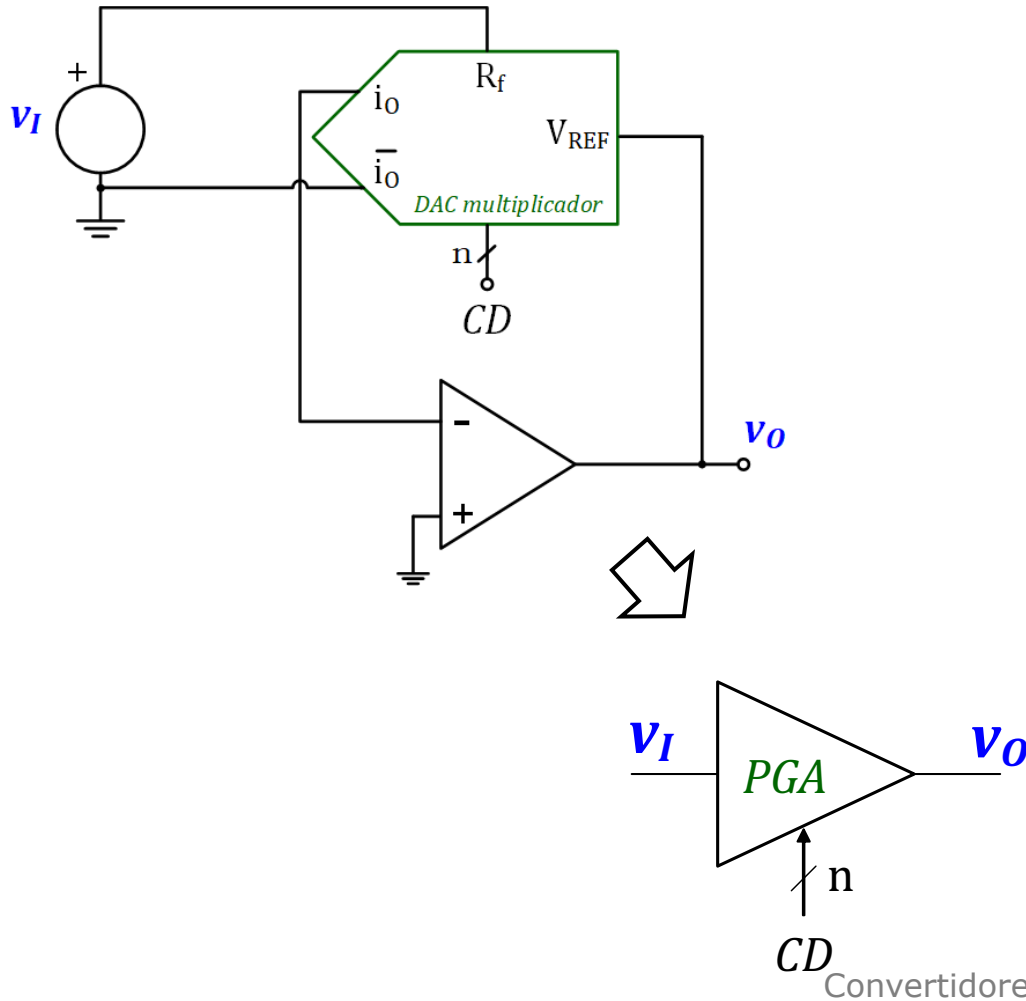
# Ejemplo de aplicación: DAC de tipo multiplicador.

El circuito de la figura es un *atenuador programable digitalmente*. Obtenga la expresión:  $A_v = (v_o/v_I)$



# Ejemplo de aplicación: DAC de tipo multiplicador.

El circuito de la figura es un *amplificador programable digitalmente (PGA)*. Obtenga la expresión:  $A_v = (v_o/v_I)$



# Índice

1. Introducción
2. Especificaciones estáticas
3. Especificaciones dinámicas
4. Técnicas de conversión
- 5. Ejemplos: DAC08 y TLC5620C**

# Ejemplo: DAC08 (la idea)



Data Sheet

8-Bit, High Speed, Multiplying  
D/A Converter

DAC08

Es un DAC de tipo multiplicador, basado en una red en escalera R-2R y fuentes de corriente ponderadas.

Su estructura interna está basada en la *siguiente idea*:

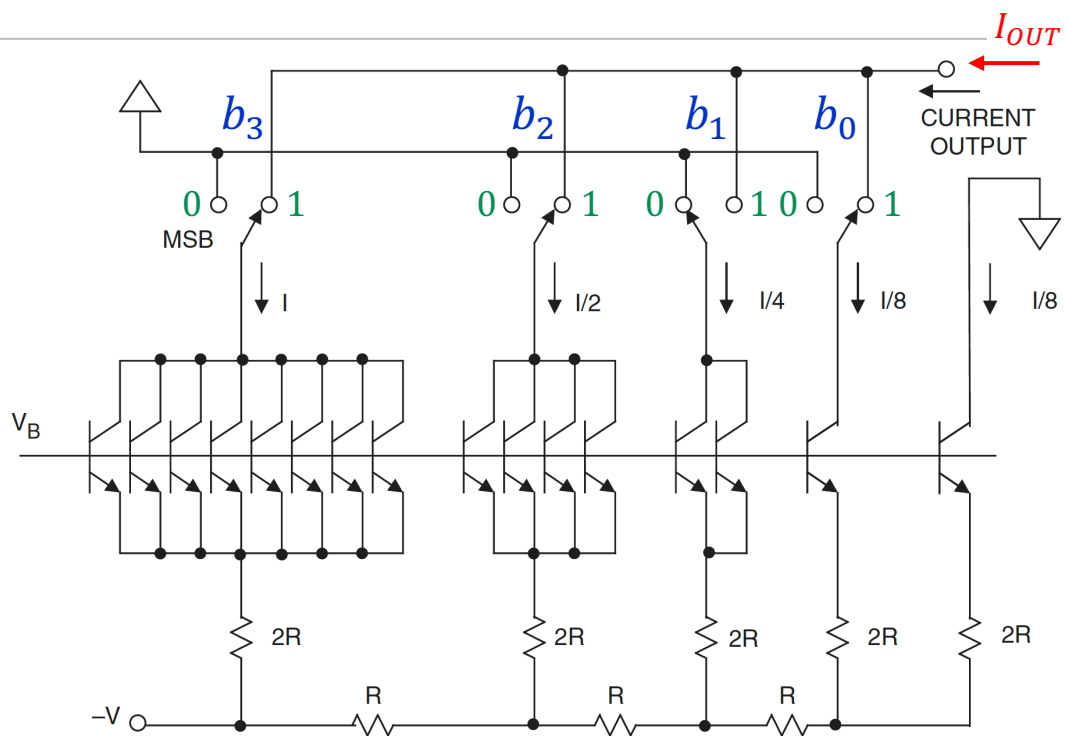


Figure 3.21: Binary-Weighted 4-Bit DAC: R/2R  
Ladder Network Current Setting Resistors

Transistores iguales, con igual tensión  $V_{BE}$  conducen igual corriente  $I_E$ .

$$I_{OUT} = \frac{I}{2} \cdot \frac{VDC}{2^n}$$

# Ejemplo: DAC08



8-Bit, High Speed, Multiplying  
D/A Converter

Data Sheet

DAC08

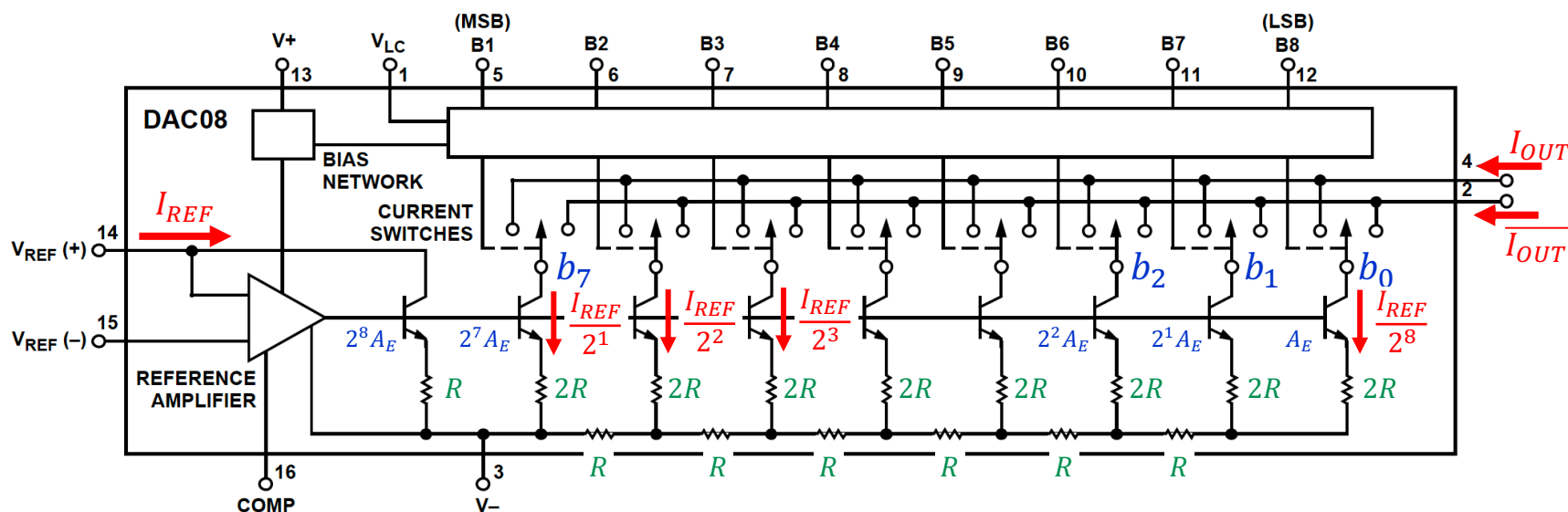
## FEATURES

- Fast settling output current: 85 ns
- Full-scale current prematched to  $\pm 1$  LSB
- Direct interface to TTL, CMOS, ECL, HTL, PMOS
- Nonlinearity to 0.1% maximum over temperature range
- High output impedance and compliance:  $-10$  V to  $+18$  V
- Complementary current outputs
- Wide range multiplying capability: 1 MHz bandwidth
- Low FS current drift:  $\pm 10$  ppm/ $^{\circ}\text{C}$
- Wide power supply range:  $\pm 4.5$  V to  $\pm 18$  V
- Low power consumption: 33 mW at  $\pm 5$  V
- Low cost

$$I_{OUT} = I_{REF} \cdot \frac{VDC}{2^8}$$

$$\overline{I_{OUT}} = I_{REF} \cdot \frac{\overline{VDC}}{2^8}$$

## FUNCTIONAL BLOCK DIAGRAM



# Ejemplo: DAC08

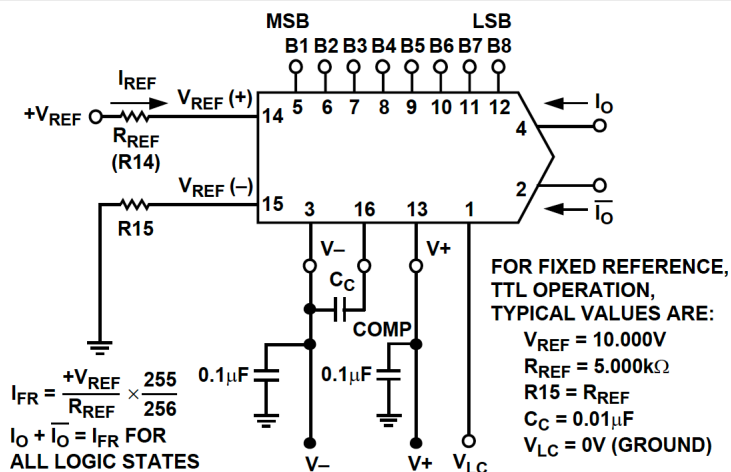
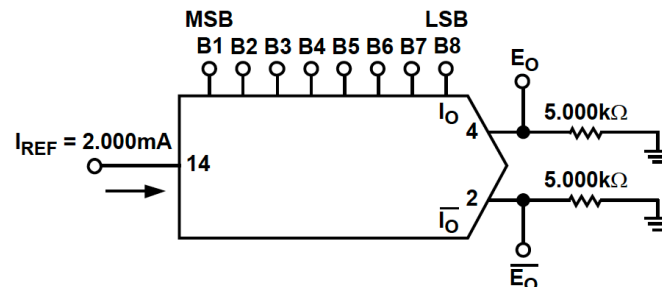
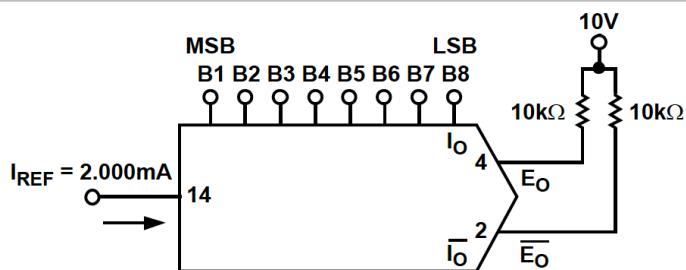


Figure 24. Basic Positive Reference Operation



|                 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | B8 | $I_O$ | $\bar{I}_O$ | $E_O$  | $\bar{E}_O$ |
|-----------------|----|----|----|----|----|----|----|----|-------|-------------|--------|-------------|
| FULL RANGE      | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1.992 | 0.000       | -9.960 | -0.000      |
| HALF SCALE +LSB | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 1.008 | 0.984       | -5.040 | -4.920      |
| HALF SCALE      | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1.000 | 0.992       | -5.000 | -4.960      |
| HALF SCALE -LSB | 0  | 1  | 1  | 1  | 0  | 1  | 1  | 1  | 0.992 | 1.000       | -4.960 | -5.000      |
| ZERO SCALE +LSB | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0.008 | 1.984       | -0.040 | -9.920      |
| ZERO SCALE      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0.000 | 1.992       | 0.000  | -9.960      |

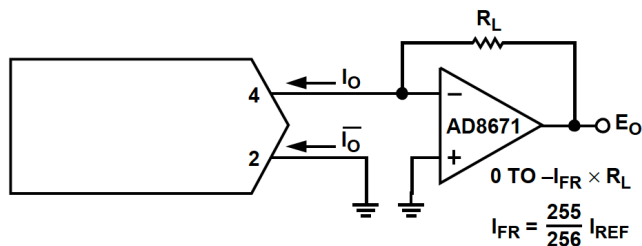
Figure 25. Basic Unipolar Negative Operation



|                      | B1 | B2 | B3 | B4 | B5 | B6 | B7 | B8 | $E_O$   | $\bar{E}_O$ |
|----------------------|----|----|----|----|----|----|----|----|---------|-------------|
| POS. FULL RANGE      | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | -9.920  | +10.000     |
| POS. FULL RANGE -LSB | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0  | -9.840  | +9.920      |
| ZERO SCALE +LSB      | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | -0.080  | +0.160      |
| ZERO SCALE           | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0.000   | +0.080      |
| ZERO SCALE -LSB      | 0  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | +0.080  | 0.000       |
| NEG. FULL SCALE +LSB | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | +9.920  | -9.840      |
| NEG. FULL SCALE      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | +10.000 | -9.920      |

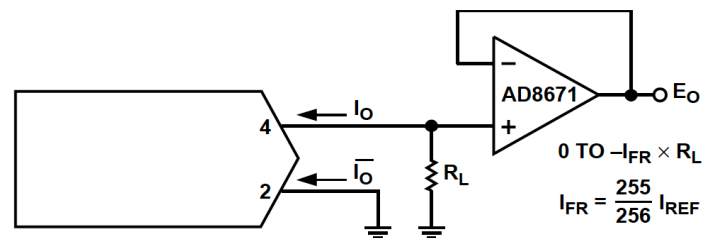
Figure 26. Basic Bipolar Output Operation

# Ejemplo: DAC08



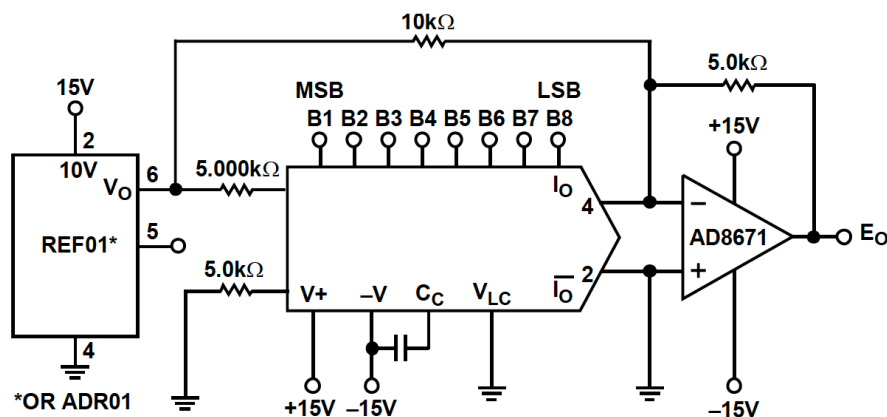
FOR COMPLEMENTARY OUTPUT (OPERATION AS A NEGATIVE LOGIC DAC).  
CONNECT INVERTING INPUT OF OP AMP TO  $\bar{I}_O$  (PIN 2): CONNECT  $I_O$  (PIN 4)  
TO GROUND.

Figure 30. Positive Low Impedance Output Operation



FOR COMPLEMENTARY OUTPUT (OPERATION AS A NEGATIVE LOGIC DAC).  
CONNECT NONINVERTING INPUT OF OP AMP TO  $\bar{I}_O$  (PIN 2): CONNECT  $I_O$  (PIN 4)  
TO GROUND.

Figure 31. Negative Low Impedance Output Operation



|                       | B1 | B2 | B3 | B4 | B5 | B6 | B7 | B8 | E <sub>O</sub> |
|-----------------------|----|----|----|----|----|----|----|----|----------------|
| POS. FULL RANGE       | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | +4.960         |
| ZERO SCALE            | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0.000          |
| NEG. FULL SCALE +1LSB | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | -4.960         |
| NEG. FULL SCALE       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | -5.000         |

Figure 29. Offset Binary Operation



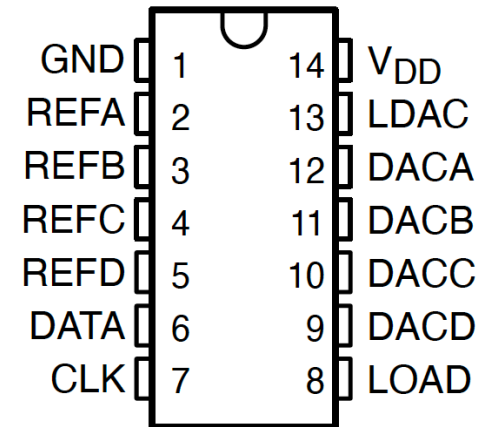
# Ejemplo: TLC5620C

- Four 8-Bit Voltage Output DACs
- 5-V Single-Supply Operation
- Serial Interface
- High-Impedance Reference Inputs
- Programmable 1 or 2 Times Output Range
- Simultaneous Update Facility
- Internal Power-On Reset
- Low-Power Consumption
- Half-Buffered Output

## applications

- Programmable Voltage Sources
- Digitally Controlled Amplifiers/Attenuators
- Mobile Communications
- Automatic Test Equipment
- Process Monitoring and Control
- Signal Synthesis

**N OR D PACKAGE  
(TOP VIEW)**

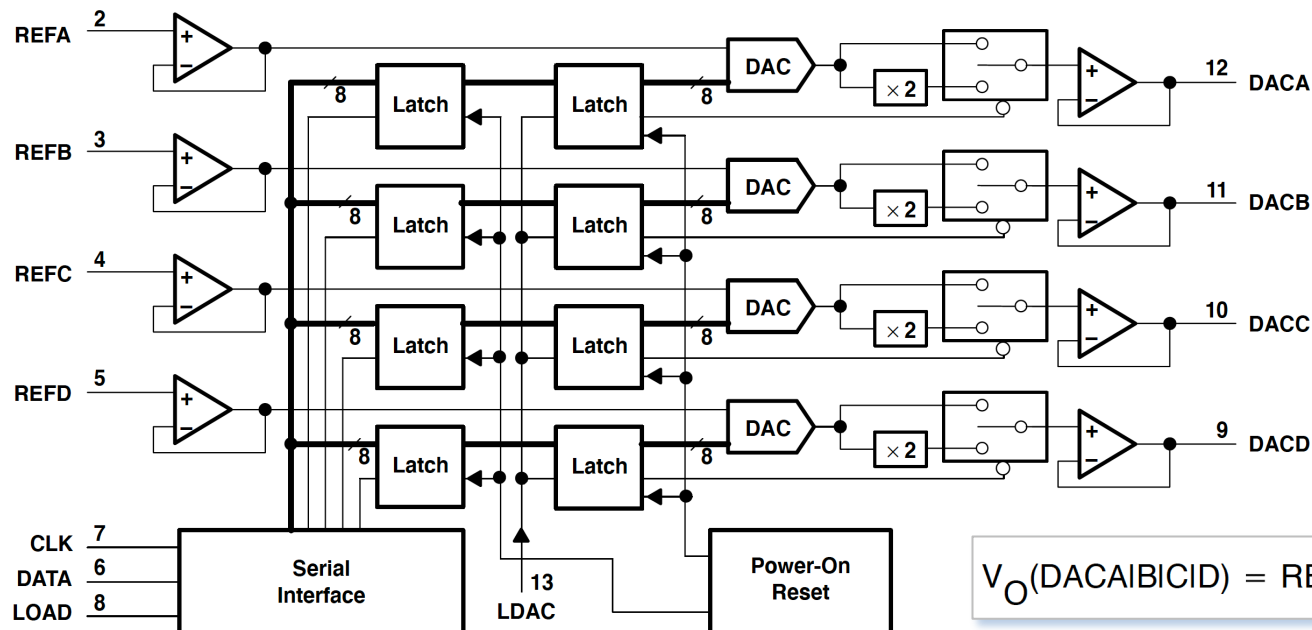


# Ejemplo: TLC5620C

## TLC5620C, TLC5620I QUADRUPLÉ 8-BIT DIGITAL-TO-ANALOG CONVERTERS

SLAS081E – NOVEMBER 1994 – REVISED NOVEMBER 2001

functional block diagram



$$V_O(\text{DACAIBICID}) = \text{REF} \times \frac{\text{CODE}}{256} \times (1 + \text{RNG bit value})$$

Table 2. Serial Input Decode

| A1 | A0 | DAC UPDATED |
|----|----|-------------|
| 0  | 0  | DACA        |
| 0  | 1  | DACB        |
| 1  | 0  | DACC        |
| 1  | 1  | DACD        |

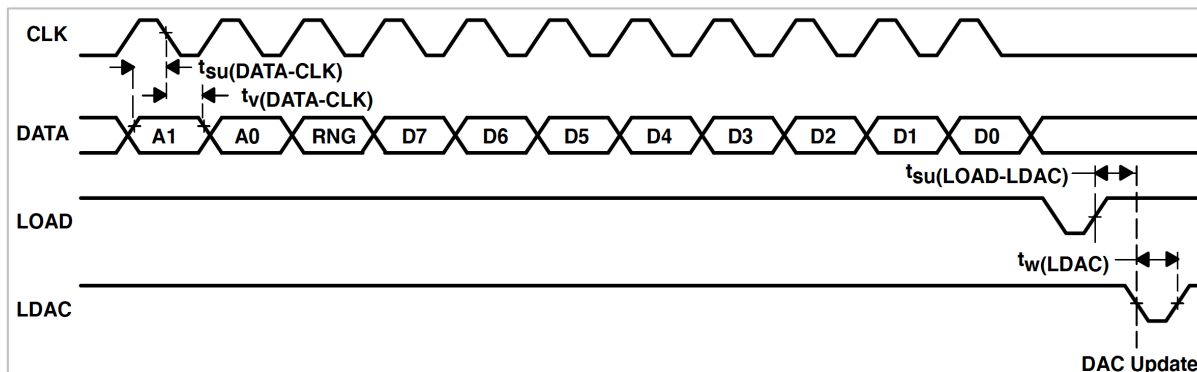


Figure 2. LDAC-Controlled Update

## Terminal Functions

| TERMINAL<br>NAME | NO. | I/O | DESCRIPTION                                                                                                                                                                                                   |
|------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLK              | 7   | I   | Serial interface clock. The input digital data is shifted into the serial interface register on the falling edge of the clock applied to the CLK terminal.                                                    |
| DACA             | 12  | O   | DAC A analog output                                                                                                                                                                                           |
| DACB             | 11  | O   | DAC B analog output                                                                                                                                                                                           |
| DACC             | 10  | O   | DAC C analog output                                                                                                                                                                                           |
| DACD             | 9   | O   | DAC D analog output                                                                                                                                                                                           |
| DATA             | 6   | I   | Serial interface digital data input. The digital code for the DAC is clocked into the serial interface register serially. Each data bit is clocked into the register on the falling edge of the clock signal. |
| GND              | 1   | I   | Ground return and reference terminal                                                                                                                                                                          |
| LDAC             | 13  | I   | Load DAC. When the LDAC signal is high, no DAC output updates occur when the input digital data is read into the serial interface. The DAC outputs are only updated when LDAC is taken from high to low.      |
| LOAD             | 8   | I   | Serial Interface load control. When LDAC is low, the falling edge of the LOAD signal latches the digital data into the output latch and immediately produces the analog voltage at the DAC output terminal.   |
| REFA             | 2   | I   | Reference voltage input to DAC A. This voltage defines the output analog range.                                                                                                                               |
| REFB             | 3   | I   | Reference voltage input to DAC B. This voltage defines the output analog range.                                                                                                                               |
| REFC             | 4   | I   | Reference voltage input to DAC C. This voltage defines the output analog range.                                                                                                                               |
| REFD             | 5   | I   | Reference voltage input to DAC D. This voltage defines the output analog range.                                                                                                                               |
| V <sub>DD</sub>  | 14  | I   | Positive supply voltage                                                                                                                                                                                       |

electrical characteristics over recommended operating free-air temperature range,  $V_{DD} = 5\text{ V} \pm 5\%$ ,  $V_{ref} = 2\text{ V}$ ,  $\times 1$  gain output range (unless otherwise noted)

| PARAMETER       |                                          | TEST CONDITIONS          |                              | MIN | TYP      | MAX       | UNIT                         |
|-----------------|------------------------------------------|--------------------------|------------------------------|-----|----------|-----------|------------------------------|
| $I_{IH}$        | High-level input current                 | $V_I = V_{DD}$           |                              |     |          | $\pm 10$  | $\mu\text{A}$                |
| $I_{IL}$        | Low-level input current                  | $V_I = 0\text{ V}$       |                              |     |          | $\pm 10$  | $\mu\text{A}$                |
| $I_{O(sink)}$   | Output sink current                      | Each DAC output          |                              | 20  |          |           | $\mu\text{A}$                |
| $I_{O(source)}$ | Output source current                    |                          |                              | 2   |          |           | $\text{mA}$                  |
| $C_i$           | Input capacitance                        |                          |                              |     | 15       |           | $\text{pF}$                  |
|                 | Reference input capacitance              |                          |                              |     | 15       |           |                              |
| $I_{DD}$        | Supply current                           | $V_{DD} = 5\text{ V}$    |                              |     |          | 2         | $\text{mA}$                  |
| $I_{ref}$       | Reference input current                  | $V_{DD} = 5\text{ V}$ ,  | $V_{ref} = 2\text{ V}$       |     |          | $\pm 10$  | $\mu\text{A}$                |
| $E_L$           | Linearity error (end point corrected)    | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 1) |     |          | $\pm 1$   | LSB                          |
| $E_D$           | Differential-linearity error             | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 2) |     |          | $\pm 0.9$ | LSB                          |
| $E_{ZS}$        | Zero-scale error                         | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 3) | 0   |          | 30        | $\text{mV}$                  |
|                 | Zero-scale-error temperature coefficient | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 4) |     | 10       |           | $\mu\text{V}/^\circ\text{C}$ |
| $E_{FS}$        | Full-scale error                         | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 5) |     |          | $\pm 60$  | $\text{mV}$                  |
|                 | Full-scale-error temperature coefficient | $V_{ref} = 2\text{ V}$ , | $\times 2$ gain (see Note 6) |     | $\pm 25$ |           | $\mu\text{V}/^\circ\text{C}$ |
| PSRR            | Power-supply rejection ratio             | See Notes 7 and 8        |                              |     | 0.5      |           | $\text{mV/V}$                |

- NOTES:
1. Integral nonlinearity (INL) is the maximum deviation of the output from the line between zero and full scale (excluding the effects of zero code and full-scale errors).
  2. Differential nonlinearity (DNL) is the difference between the measured and ideal 1 LSB amplitude change of any two adjacent codes. Monotonic means the output voltage changes in the same direction (or remains constant) as a change in the digital input code.
  3. Zero-scale error is the deviation from zero voltage output when the digital input code is zero.
  4. Zero-scale-error temperature coefficient is given by:  $ZSETC = [ZSE(T_{max}) - ZSE(T_{min})]/V_{ref} \times 10^6/(T_{max} - T_{min})$ .
  5. Full-scale error is the deviation from the ideal full-scale output ( $V_{ref} - 1\text{ LSB}$ ) with an output load of  $10\text{ k}\Omega$ .
  6. Full-scale-error temperature coefficient is given by:  $FSETC = [FSE(T_{max}) - FSE(T_{min})]/V_{ref} \times 10^6/(T_{max} - T_{min})$ .
  7. Zero-scale-error rejection ratio (ZSE RR) is measured by varying the  $V_{DD}$  from  $4.5\text{ V}$  to  $5.5\text{ V}$  dc and measuring the proportion of this signal imposed on the zero-code output voltage.
  8. Full-scale-error rejection ratio (FSE RR) is measured by varying the  $V_{DD}$  from  $4.5\text{ V}$  to  $5.5\text{ V}$  dc and measuring the proportion of this signal imposed on the full-scale output voltage.

**operating characteristics over recommended operating free-air temperature range,  $V_{DD} = 5\text{ V} \pm 5\%$ ,  $V_{ref} = 2\text{ V}$ ,  $\times 1$  gain output range (unless otherwise noted)**

|                              | TEST CONDITIONS                                                                          | MIN | TYP | MAX | UNIT             |
|------------------------------|------------------------------------------------------------------------------------------|-----|-----|-----|------------------|
| Output slew rate             | $C_L = 100\text{ pF}$ , $R_L = 10\text{ k}\Omega$                                        |     | 1   |     | V/ $\mu\text{s}$ |
| Output settling time         | To $\pm 0.5\text{ LSB}$ , $C_L = 100\text{ pF}$ , $R_L = 10\text{ k}\Omega$ , See Note 9 |     | 10  |     | $\mu\text{s}$    |
| Large-signal bandwidth       | Measured at $-3\text{ dB}$ point                                                         |     | 100 |     | kHz              |
| Digital crosstalk            | CLK = 1-MHz square wave measured at DACA-DACD                                            |     | -50 |     | dB               |
| Reference feedthrough        | See Note 10                                                                              |     | -60 |     | dB               |
| Channel-to-channel isolation | See Note 11                                                                              |     | -60 |     | dB               |
| Reference input bandwidth    | See Note 12                                                                              |     | 100 |     | kHz              |

- NOTES: 9. Settling time is the time between a LOAD falling edge and the DAC output reaching full scale voltage within  $\pm 0.5\text{ LSB}$  starting from an initial output voltage equal to zero.
10. Reference feedthrough is measured at any DAC output with an input code = 00 hex with a  $V_{ref}$  input =  $1\text{ V dc} + 1\text{ V}_{pp}$  at 10 kHz.
11. Channel-to-channel isolation is measured by setting the input code of one DAC to FF hex and the code of all other DACs to 00 hex with  $V_{ref}$  input =  $1\text{ V dc} + 1\text{ V}_{pp}$  at 10 kHz.
12. Reference bandwidth is the  $-3\text{ dB}$  bandwidth with an input at  $V_{ref} = 1.25\text{ V dc} + 2\text{ V}_{pp}$  and with a full-scale digital-input code.